

YENEPPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE AND MANAGEMENT
A CONSTITUENT UNIT OF YENEPPOYA (DEEMED TO BE UNIVERSITY)
BALMATTA, MANGALORE

INTERNSHIP REPORT ON
DATA ANALYTICS
AUSTRIX TECHNOLOGIES LLC

SUBMITTED BY
SAYYED MUHAMMED SHAMMAS
III BCA
(DATA SCIENCE AND BIG DATA ANALYTICS)
23BCDSBA76

UNDER THE GUIDANCE OF
MS. BHOOMIKA SUVARNA
LECTURER
DEPARTMENT OF COMPUTER SCIENCE

IN PARTIAL FULLFILLMENT OF THE REQUIREMENT
FOR THE AWARD OF THE DEGREE OF

BACHELOR OF COMPUTER APPLICATIONS
(DATA SCIENCE AND BIG DATA ANALYTICS)
MAY -2026

INTERNSHIP REPORT

ON

DATA ANALYTICS

Internship organization



Austrix Technologies LLC

Middle East | Africa | India

Web: www.auStrix.com, Email: mailus@auStrix.com

Under the guidance of

Ms. Bhoomika Suvarna

Lecturer

Department of Computer Science

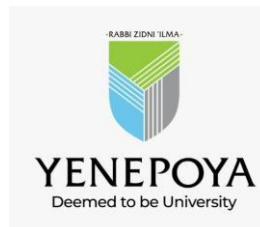
Submitted by

Sayyed Muhammed Shammas

23BCDSBA76

BACHELOR OF COMPUTER APPLICATIONS

(Data Science and Big Data Analytics)



YENEPOYA INSTITUTE OF ARTS, SCIENCE, COMMERCE & MANAGEMENT

(A constituent unit of Yenepoya Deemed to be University)

Deralakatte, Mangaluru – 575018, Karnataka, India

CERTIFICATE

This is to certify that the project work entitled “**Data Science and Big Data Analytics**” has been successfully carried out at “**Austrix Technologies LLC**” by **SAYYED MUHAMMED SHAMMAS** (Reg. No. 23BCDSBA76), student of **3rd Year BCA (Data Science and Big Data Analytics)** under the supervision and guidance of **Ms. Bhoomika Suvarna**, Lecturer, Department of Computer Science, **Yenepoya Institute of Arts, Science, Commerce and Management**. This dissertation is submitted in partial fulfilment for the award of degree in **Bachelor of Computer Applications** by **Yenepoya (Deemed to be university)** during academic year 2025-26.

Internal Guide: Ms. Bhoomika Suvarna

Chairperson

Internal Examiner:

External Examiner

PRINCIPAL

Dr. Jeevan Raj

**The Yenepoya Institute of Arts,
Science, Commerce and Management**
(Deemed To be university)

Submitted for the viva-voice

Place: Mangalore

Examination held on:

Date

DECLARATION

I, **SAYYED MUHAMMED SHAMMAS** a student of **BCA (Data Science and Big Data Analytics)** at **Yenepoya Institute of Arts, Science, Commerce, and Management, Balmatta, Mangalore**, affiliated with **Yenepoya (Deemed to be University)**, hereby declare that this internship report titled **Coffee Shop Sales** is a genuine and original record of the work undertaken by me as part of my academic curriculum.

This report documents the **knowledge, skills, and practical experience** acquired during my internship at **Austrix Technologies LLC**. It includes methodologies, analytical processes, and investigative approaches aligned with **recognized industry standards in digital forensics, incident response, and cybersecurity investigations**.

I extend my sincere gratitude to **Ms. Ayshath Lubna, Internship Program Head**, for her **valuable guidance, mentorship, and support** throughout the internship period. I also express my appreciation to **Austrix Technologies LLC** for providing an enriching environment and exposure to real-world cybersecurity operations.

Furthermore, I acknowledge the support and encouragement received from my institutional guide, **Ms. Bhoomika Suvarna**, and the faculty of **Yenepoya Institute of Arts, Science, Commerce, and Management**. Their insights and assistance have been instrumental in successfully completing this internship.

Intern Signature:

Name: Sayyed Muhammed shammas

Date:

Institution Guide Signature:

Name: Ms. Bhoomika Suvarna

Institute: Yenepoya Institute of Arts, Science, Commerce, and Management

Date:

Internship Supervisor Signature:

Name: Ms. Ayshath Lubna

Designation: Internship Program Head

Organization: Austrix Technologies LLC

Date:

Head of the Department (HOD) Signature:

Name: Dr. Rathnakar Shetty P

Head of Department of Computer Science

Institute: Yenepoya Institute of Arts, Science, Commerce, and Management

Date:

ACKNOWLEDGEMENT

I sincerely express my gratitude to **Yenepoya Institute of Arts, Science, Commerce, and Management, affiliated with Yenepoya (Deemed to be University), Mangalore**, for providing me with the opportunity to undertake this internship as part of my academic curriculum.

I extend my heartfelt thanks to **Dr. Jeevan Raj, Principal**, for his continuous support and guidance in facilitating this learning opportunity. I also express my sincere appreciation to **Dr. Rathnakar Shetty P**, Head of Department of Computer Science, for his encouragement and academic support throughout my internship journey.

I am deeply grateful to **Ms. Ayshath Lubna** for her visionary leadership and for fostering a dynamic learning environment in **Austrix Technologies LLC**. A special note of appreciation goes to **Ms. Ayshath Lubna** for her invaluable mentorship, expert guidance, and for sharing her vast experience in the field.

I also extend my sincere gratitude to **Ms. Ayshath Lubna** for her technical insights, hands-on training, and unwavering support during my internship. Their guidance has been instrumental in enhancing my practical understanding of cyber forensics and incident response methodologies.

I am also thankful to my internal guide, **Ms. Bhoomika Suvarna**, for her academic mentorship and for providing valuable insights that helped bridge the gap between theoretical learning and practical application.

Lastly, I am grateful to my faculty mentors, family, and peers for their encouragement and motivation throughout this journey.

Name: Sayyed Muhammed shammass

Reg NO: 23BCDSBA76

Date:

INTERNSHIP OFFER LETTER



auStrix Technologies LLC
Middle East | Africa | India
Web: www.auStrix.com
Email: mail.in@austrix.com

Date: February 19, 2026

To,

Sayyed Muhammed Shammās

Email : chemmuchaa2@gmail.com

Phone No : 7356223651

The Yenepoya Institute of Arts, Science, Commerce & Management, Mangalore

Bachelor of Computer Applications (BCA)

Academic Year : 2023-2026

Subject : Internship Offer Letter

Dear Sayyed Muhammed Shammās,

We are pleased to offer you an internship position at **Austrix Technologies LLC**. Your internship will be for a period of **60 days**, starting from **March 1, 2026, to May 1, 2026**. During this period, you will work closely with our team to gain practical experience in Data Analytics.

Internship Details:

Position: Data Analyst Intern

Duration: 60 days

Location: Austrix Technologies LLC, Kasaragod

Roles & Responsibilities

During this internship, you will be expected to:

- Data cleaning and preprocessing
- Perform exploratory data analysis (EDA)
- Create dashboards and reports using tools such as Excel / SQL / Python / Power BI
- Assist in building data models and generating insights
- Utilize advanced AI tools.
- Present findings to the reporting supervisor

Upon successful completion of the internship and assigned projects, you will receive an **Internship Completion Certificate** from **Austrix Technologies LLC**.

Kindly confirm your acceptance of this offer by replying to this email before **23-02-2026**. We look forward to having you join our team and wish you a productive learning experience.

Best Regards,

Shabeer B

Manager, Austrix Technologies LLC

shabeer@austrix.com

Mobile: +91 7592878749

INTERNSHIP COMPLETION CERTIFICATE



COMPANY PROFILE: [Austrix Technologies]



Website: www.austrix.com Founded: 1999

Headquarters: Canada

Support Centers: USA, UAE, India, Qatar

Overview:

Austrix Technologies is a global provider of enterprise-grade web hosting and IT infrastructure solutions. With over two decades of experience, the company delivers scalable, secure, and high-performance hosting environments tailored for businesses of all sizes.

Core Services:

- Web & Email Hosting – High-speed, secure hosting with spam protection, DKIM support, and mailbox migration tools.
- Domain Registration – Global domain registration with DNS management.
- Cloud Solutions & Dedicated Servers – Reliable, scalable hosting infrastructure with 99.9% uptime.
- Custom Solutions – Digital signage, enterprise mail services, and application hosting customized for business needs.

Key Features:

- 24/7 Global Support
- Data Redundancy & Daily Backups
- Real-time Monitoring & Performance Dashboards
- Multi-layer Security (network, host, application)
- Disaster Recovery & RAID Technology

Why Austrix:

- Proven track record of reliable service delivery
- Infrastructure spread across continents
- Strong emphasis on data protection and uptime

Contact: Visit: www.austrix.com/contact-us.aspx



ROLES AND RESPONSIBILITIES

During the internship, I was actively involved in performing end-to-end data analysis and dashboard development on coffee shop sales data using the Coffee Shop Sales dataset. My key roles and responsibilities included:

Data Collection and Understanding

- Imported and worked with the Coffee Shop Sales dataset using Python (Pandas)
- Explored dataset structure using functions like `.info()`, `.head()`, and `.describe()`
- Identified important features such as Transaction Date, Transaction Time, Store Location, Product Category, Product Type, Product Detail, Transaction Quantity, and Unit Price
- Understood the business significance of each variable to support effective analysis and business decision-making

Data Cleaning and Preprocessing

- Checked for missing values using `.isnull().sum()` and ensured data consistency
- Converted Transaction Date and Transaction Time into proper datetime formats
- Verified and corrected data types for accurate analysis
- Removed duplicate records and handled inconsistencies to maintain clean and reliable data
- Created a new feature called Total Sales using the formula:
- **Total Sales = Transaction Quantity × Unit Price**
- Extracted additional time-based features such as Month, Day, Weekday, Hour, and Weekend/Weekday indicators for advanced analysis

Exploratory Data Analysis (EDA)

- Analyzed sales performance across different store locations
- Identified top-performing products based on total revenue and quantity sold
- Compared sales across product categories such as Coffee, Tea, Bakery, Chocolate, and other beverages
- Studied customer purchasing behavior during weekdays and weekends
- Evaluated hourly sales trends to identify peak business hours and customer traffic patterns

Data Visualization Using Python

- Created visualizations using Matplotlib and Seaborn for better interpretation of insights
- Developed:
 - Bar charts for store-wise sales, product categories, and monthly performance
 - Line charts for daily and hourly sales trends
 - Pie charts for category-wise revenue contribution
 - Heatmaps to analyze weekday vs hourly sales performance
 - Scatter plots to study relationships between unit price and quantity sold

Power BI Dashboard Development

- Designed and developed an interactive single-page Power BI dashboard for comprehensive business analysis
- Created KPI cards to display:
 - Total Sales
 - Total Orders
 - Total Quantity Sold
 - Average Sales
- Built interactive visualizations including:
 - Monthly Sales Trend Analysis
 - Sales by Product Category
 - Store Location Performance
 - Hourly Sales Analysis
 - Top Selling Products
 - Weekday vs Weekend Sales Comparison
- Added slicers and filters for:
 - Month
 - Store Location
 - Product Category
 - Weekday/Weekend analysis
- Used DAX measures to calculate key business metrics and improve dashboard interactivity
- Designed the dashboard using a professional coffee-shop-themed layout for better visual presentation and business reporting

Trend and Relationship Analysis

- Analyzed monthly sales trends to identify seasonal growth patterns
- Compared store-wise performance differences in revenue generation
- Studied the relationship between product pricing and sales quantity
- Evaluated the impact of weekdays, weekends, and peak morning hours on revenue
- Identified product categories and product types contributing most to total sales

Insight Generation and Reporting

- Identified key business insights such as the dominance of Coffee and Tea in overall sales
- Highlighted top-performing stores, products, and categories
- Explained how peak business hours significantly influence revenue generation
- Analyzed customer purchasing patterns across different days and timings
- Prepared structured reports, dashboards, and visual presentations to support business understanding and data-driven decision-making

LEARNING OUTCOMES

This internship provided a valuable opportunity to apply theoretical knowledge of data analytics and business intelligence to a real-world coffee shop sales dataset. Working on the Coffee Shop Sales Analysis project helped me understand how raw transactional data can be transformed into meaningful insights that explain sales performance, customer purchasing behavior, product demand, and store-wise revenue trends.

Throughout this project, I gained both technical and analytical skills, along with a deeper understanding of how data-driven insights and interactive dashboards support effective business decision-making.

1. Understanding the Importance of Data Cleaning and Preparation

One of the most important lessons I learned during this project was the significance of data cleaning and preprocessing. Although the dataset was well-structured, it still required careful handling to ensure accuracy and consistency.

For example, columns such as Transaction Date and Transaction Time needed to be converted into proper datetime format for time-based analysis. I also checked for missing values and duplicate records to ensure data reliability.

Additionally, I verified that numerical columns such as Transaction Quantity and Unit Price were correctly formatted for calculations. I created a new column called Total Sales using the formula:

$$\text{Total Sales} = \text{Transaction Quantity} \times \text{Unit Price}$$

I also extracted additional features such as Month, Day, Weekday, Hour, and Weekend/Weekday indicators for advanced time-series analysis.

This experience taught me that data preprocessing is a critical step because inaccurate or unclean data can lead to misleading analytical results and poor business decisions.

2. Developing Strong Exploratory Data Analysis (EDA) Skills

This project helped me build a strong foundation in Exploratory Data Analysis (EDA), which is essential for identifying hidden patterns and trends in business data.

I explored the dataset by:

- Analyzing sales performance across different store locations
- Identifying top-performing products based on total revenue
- Comparing performance across product categories such as Coffee, Tea, Bakery, and Drinking Chocolate
- Studying customer purchasing behavior during weekdays and weekends
- Evaluating peak business hours and customer traffic patterns

Through this process, I discovered that Coffee and Tea contribute the highest share of revenue and that morning hours generate the maximum sales due to breakfast and office-hour demand.

EDA helped me understand how to explore datasets effectively, ask meaningful business questions, and generate valuable insights for decision-making.

3. Data Visualization and Interpretation

Another major learning outcome was the ability to visually represent data using Matplotlib, Seaborn, and Power BI.

Using Python libraries, I created:

- Bar charts to compare sales across stores, products, and categories
- Line charts to analyze monthly, daily, and hourly sales trends
- Pie charts to represent category-wise revenue contribution
- Heatmaps to study weekday and hourly sales patterns
- Scatter plots to analyze relationships between unit price and quantity sold

In addition to Python visualizations, I designed an interactive single-page Power BI dashboard to present the analysis in a professional and user-friendly manner.

The Power BI dashboard included:

- KPI cards for Total Sales, Total Orders, Total Quantity Sold, and Average Sales
- Monthly Sales Trend Analysis
- Sales by Product Category
- Store Location Performance Analysis
- Hourly Sales Analysis
- Top Selling Products Visualization
- Weekday vs Weekend Sales Comparison
- Interactive slicers for Month, Store Location, Product Category, and Weekday analysis

This dashboard helped convert raw data into interactive business insights and improved my understanding of business intelligence tools and dashboard design principles.

I learned that effective visualization and dashboarding are essential for storytelling with data and for communicating insights clearly to both technical and non-technical audiences.

4. Understanding Relationships Between Variables

One of the most valuable insights from this project was understanding the relationship between product pricing and quantity sold.

Through analysis and visualization, I observed that lower-priced products were sold in higher quantities, while higher-priced items were purchased less frequently. This demonstrated how pricing directly influences customer purchasing behavior.

I also analyzed monthly sales trends and identified steady sales growth during specific months such as May and June.

Additionally, I studied how factors such as:

- Product type
- Store location
- Purchase timing
- Weekdays and weekends

This improved my understanding of relationship analysis and trend evaluation in real-world business datasets.

5. Analytical Thinking and Insight Generation

This project significantly improved my analytical thinking and problem-solving skills.

Instead of simply creating charts and dashboards, I learned how to:

- Interpret data logically
- Connect insights to real-world business scenarios
- Identify patterns and explain customer behavior
- Generate actionable business recommendations

For example:

- Higher sales during morning hours indicate strong customer demand for coffee during breakfast and office timings
- Weekday sales are higher because regular working-day customers contribute significantly to daily revenue
- Certain product categories consistently outperform others in terms of sales and customer demand

This experience strengthened my ability to transform raw data into meaningful business insights that can support strategic decision-making.

6. Communication and Presentation Skills

An important part of this internship was presenting findings in a clear, structured, and professional manner.

I learned how to:

- Organize analytical reports logically
- Design interactive dashboards for business presentation
- Write conclusions in a simple and understandable format
- Explain technical findings to non-technical audiences

Preparing reports, visualizations, and dashboard presentations improved my confidence in presenting my work during project reviews and viva sessions.

Overall, this internship was a highly valuable learning experience that enhanced my technical, analytical, and presentation skills. Working on the Coffee Shop Sales Analysis project helped me gain practical exposure to data cleaning, exploratory data analysis, visualization, dashboard development, and business intelligence reporting using Python and Power BI.

The project also strengthened my understanding of how data analytics can support real-world business operations and improve data-driven decision-making in the retail and food service industry.

WEEKLY INTERNSHIP LOG

INTERN WEEKLY REPORT

Intern: Sayed Mubammad Shamir Week Beginning: 02/03/26 To: 05/03/26

Position: Date Analytics Intern

Employer: Austrix Technologies

Internship Supervisor's Name: Aysbeth Lubin Phone: 9946856681

Internship Coordinator: Aysbeth Lubin

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	10:00	1:00	3	Program Introduction & Admissions
TUESDAY	1:00	4:00	3	Introduction to Excel and Basic operations
WEDNESDAY	1:00	4:00	3	Date Handling and date Validation
THURSDAY	1:00	4:00	3	Logical Functions and Lookup Functions
FRIDAY				
SATURDAY				
SUNDAY				
TOTAL HOURS				

Signatures:

Intern

Internship Supervisor

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INTERN WEEKLY REPORT

Intern: Sayed Nuhman Shams Week Beginning: 01/03/26 To: 11/03/26

Position: Data Analytics Intern

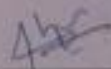
Employer: Aushix Technologies

Internship Supervisor's Name: Aisheth Lubna Phone: 9946886681

Internship Coordinator: Aisheth Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	1:00	4:00	3	Advanced Excel Functions and Power Query
TUESDAY	1:00	4:00	3	Power pivot & Power Tables
WEDNESDAY	1:00	4:00	3	Power charts, Dashboarding and Final project
THURSDAY				
FRIDAY				
SATURDAY				
SUNDAY				
TOTAL HOURS				

Signatures: 
Intern


Internship Supervisor

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INTERN WEEKLY REPORT

Intern: Sayed Muhammad Shams Week Beginning: 16-3-2026 To: 18-03-2026

Position:

Data Analytics Intern

Employer:

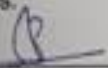
Austrix Technologies

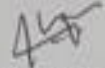
Internship Supervisor's Name: Ayshath Lubna Phone: 9946886681

Internship Coordinator: Ayshath Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	10.00	2.00	4	Excel Project Work (Part 1)
TUESDAY	10.00	2.00	4	Excel Project Work (Part 2)
WEDNESDAY	10.00	2.00	4	Introduction to SQL & Database Concepts
THURSDAY				
FRIDAY				
SATURDAY				
SUNDAY				
TOTAL HOURS			12	

Signatures:


Intern


Internship Supervisor

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INTERN WEEKLY REPORT

Intern: Sayyed Nuhammed Shemmer Week Beginning: 24-3-2026 To: 27-03-2026

Position:

Data Analytics Intern

Employer:

Austrix Technologies

Internship Supervisor's Name: Ayshath Lubna Phone: 9946886681

Internship Coordinator: Ayshath Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	10.00	2.00	4	DDL & Table Creation
TUESDAY	10.00	2.00	4	DML Operations & Constraints
WEDNESDAY	10.00	2.00	4	Filtering, Sorting & SQL Joins
THURSDAY				
FRIDAY				
SATURDAY				
SUNDAY				
TOTAL HOURS			12	

Signatures:


Intern


Internship Supervisor

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INTERN WEEKLY REPORT

Intern: Sayed Muhammad Shammes Week Beginning: 06-04-2026 To: 10-04-2026

Position:

Data Analytics Intern

Employer:

Austrix Technologies

Internship Supervisor's Name: Ayshath Lubna Phone: 9946886681

Internship Coordinator: Ayshath Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	10.00	2.00	4	Data Cleaning in Python
TUESDAY	10.00	2.00	4	Data Transformation, Sorting & Ranking
WEDNESDAY	10.00	2.00	4	Indexing & Selection Aggregation & GroupBy Operations
THURSDAY				Holiday
FRIDAY	10.00	2.00	4	Combining Datasets Working with Strings & Date-Time Functions
SATURDAY				
SUNDAY				
TOTAL HOURS			16	

Signatures:

Intern

Internship Supervisor

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INTERN WEEKLY REPORT

Intern: Sayed Mohamed Shemmes Week Beginning: 13-04-2026 To: 17-04-2026

Position:

Data Analytics Intern

Employer:

Austrix Technologies

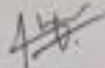
Internship Supervisor's Name: Ayshath Lubna Phone: 9946886681

Internship Coordinator: Ayshath Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	10.00	2.00	4	Exploratory Data Analysis (EDA)
TUESDAY	10.00	2.00	4	Data Visualization with Matplotlib
WEDNESDAY				Holiday
THURSDAY	10.00	2.00	4	Visualization with Seaborn
FRIDAY	10.00	2.00	4	Real Dataset Analysis
SATURDAY				
SUNDAY				
TOTAL HOURS			16	

Signatures:

Intern 

Internship Supervisor 

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INTERN WEEKLY REPORT

Intern: Sayed Muhammad Shamma Week Beginning: 20-04-2026 To: 24-04-2026

Position:

Data Analytics Intern

Employer:

Austrix Technologies

Internship Supervisor's Name: Ayshath Lubna Phone: 9946886681

Internship Coordinator: Ayshath Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY	10.00	2.00	4	Final Project Work
TUESDAY	10.00	2.00	4	Dashboard Completion
WEDNESDAY	10.00	2.00	4	Project Review & Improvements
THURSDAY	10.00	3.00	5	Final Presentation Preparation
FRIDAY	10.00	3.00	5	Project Presentation & Internship Wrap-up
SATURDAY				
SUNDAY				
TOTAL HOURS			22	

Signatures:

[Signature]
Intern

[Signature]
Internship Supervisor

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INTERN WEEKLY REPORT

Intern: Sayed Mubammad Shemmo Week Beginning: 30-3-2026 To: 03-04-2026

Position:

Data Analytics Intern

Employer:

Austrix Technologies

Internship Supervisor's Name: Ayshath Lubna Phone: 9946886681

Internship Coordinator: Ayshath Lubna

DAY	TIME IN	TIME OUT	TOTAL HOURS	DUTIES PERFORMED
MONDAY				Eid Holiday
TUESDAY	10.00	2.00	4	Aggregation, GROUP BY & HAVING
WEDNESDAY	10.00	2.00	4	SQL Mini Project
THURSDAY	10.00	2.00	4	Introduction to Python
FRIDAY	10.00	2.00	4	Pandas Introduction
SATURDAY				
SUNDAY				
TOTAL HOURS			16	

Signatures:

[Signature]
Intern

[Signature]
Internship Supervisor

PLEASE RETURN TO: Internship Guide



CERTIFICATIONS & OTHER ACTIVITIES

During the Data Analytics Internship, I successfully completed a structured and industry-oriented training program that provided practical exposure to data analysis, data preprocessing, database management, and business intelligence tools. The internship followed a systematic learning approach and enabled me to work with real-world datasets using Microsoft Excel, SQL, Python, and Power BI.

In the initial phase of the internship, I worked extensively with Microsoft Excel for data cleaning, validation, sorting, and filtering operations to prepare raw data for analysis. I applied logical functions such as IF, AND, and OR, along with lookup functions including VLOOKUP, HLOOKUP, XLOOKUP, and INDEX-MATCH to solve practical business problems. I also gained hands-on experience with Power Query and Power Pivot for data transformation and modeling. Additionally, I created interactive dashboards using Pivot Tables and Pivot Charts, which helped me understand how business data is summarized, analyzed, and presented visually for decision-making.

During the SQL phase, I developed a strong understanding of relational database concepts and query writing techniques. I worked with DDL, DML, and DQL commands to create, manage, and manipulate database tables. I performed data analysis using SQL queries involving SELECT, WHERE, ORDER BY, GROUP BY, HAVING, and aggregate functions such as SUM, COUNT, AVG, MAX, and MIN. I also applied JOIN operations to combine multiple tables and extract meaningful information from structured datasets. As part of this phase, I completed a mini project focused on generating business insights using SQL-based data analysis.

In the Python phase, I worked with libraries such as Pandas and NumPy for data preprocessing, transformation, and analysis. I handled missing values, removed duplicate records, corrected data types, and performed indexing, filtering, grouping, and aggregation operations. I also worked with string manipulation functions, date-time operations, dataset merging, and feature extraction techniques, which improved my understanding of preparing real-world datasets for analysis.

Further, I applied Exploratory Data Analysis (EDA) techniques to identify trends, patterns, and relationships within data. Using Matplotlib and Seaborn, I created various visualizations including:

- Bar charts
- Line plots
- Scatter plots
- Pie charts
- Heatmaps

These visualizations helped me improve my ability to interpret analytical results and communicate insights effectively through graphical representations.

As part of the internship, I completed a final end-to-end project on Coffee Shop Sales Analysis, where I applied all the concepts learned throughout the program. The project involved importing and preprocessing transactional sales data, creating new features such as Month, Weekday, Hour, and Weekend indicators, and analyzing important business metrics related to sales performance and customer purchasing behavior.

I analyzed:

- Monthly and daily sales trends
- Store-wise revenue performance
- Product category and product type performance
- Peak business hours
- Weekday vs weekend sales patterns
- Customer purchasing behavior
- Relationship between product pricing and quantity sold

In addition to Python-based analysis, I designed and developed an interactive Power BI Dashboard to present business insights in a visually appealing and user-friendly manner. The dashboard included:

- KPI cards for Total Sales, Total Orders, Total Quantity Sold, and Average Sales
- Monthly Sales Trend Analysis
- Product Category-wise Sales Distribution
- Store-wise Performance Analysis

- Hourly Sales Analysis
- Top Selling Products Visualization
- Weekday vs Weekend Sales Comparison
- Interactive slicers and filters for dynamic analysis

The dashboard helped in presenting complex analytical insights in a simple and understandable format, enabling better business interpretation and decision-making.

Through this project, I identified important business insights such as:

- Coffee and Tea products contributed the highest sales revenue
- Morning hours generated peak customer traffic and sales
- Certain store locations consistently outperformed others
- Weekday sales were higher compared to weekend sales
- Lower-priced products were sold in higher quantities compared to premium-priced products

Overall, this internship significantly strengthened my technical, analytical, and problem-solving skills. It improved my ability to work with real-world datasets, perform end-to-end data analysis, build interactive dashboards, and communicate insights effectively in a structured and professional manner. The experience also enhanced my confidence in using modern data analytics tools and applying data-driven approaches to solve business problems.

ABSTRACT

In today's data-driven business environment, analyzing sales performance and customer purchasing behavior plays a vital role in improving operational efficiency, customer satisfaction, and strategic decision-making. This project focuses on performing a comprehensive analysis of a Coffee Shop Sales dataset to identify trends in sales, customer behavior, product performance, and store-wise revenue generation using data analytics and business intelligence techniques.

The dataset used in this project contains transactional information such as transaction date, transaction time, store location, product category, product type, product details, transaction quantity, and unit price. Using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, the dataset was cleaned, processed, and analyzed to extract meaningful insights from raw sales data. Data preprocessing techniques including handling missing values, correcting data types, removing duplicates, and feature engineering were performed to prepare the dataset for analysis.

Additional features such as Month, Day, Weekday, Hour, and Weekend/Weekday indicators were created to support detailed time-based analysis. Various analytical techniques such as grouping, aggregation, correlation analysis, and exploratory data analysis (EDA) were applied to study customer purchasing patterns and business performance.

The project also involved the development of an interactive Power BI Dashboard to visually represent important business insights in a clear and user-friendly format. The dashboard included KPI cards for Total Sales, Total Orders, Total Quantity Sold, and Average Sales, along with visualizations such as monthly sales trends, product category analysis, hourly sales analysis, top-selling products, store-wise revenue comparison, and weekday vs weekend sales analysis. Interactive slicers and filters were used to improve dashboard usability and support dynamic data exploration.

The analysis revealed several valuable insights. Coffee and Tea products contributed the highest share of total sales revenue, while products such as Sustainably Grown Organic Large and Dark Chocolate Large were identified as top-performing products. Store locations including Hell's Kitchen, Astoria, and Lower Manhattan showed consistent sales performance and strong revenue contribution.

Time-based analysis highlighted that peak sales occurred during morning hours between 8 AM and 10 AM, indicating high customer demand during breakfast and office hours. The analysis also showed that weekday sales were generally higher than weekend sales, suggesting a strong contribution from regular working-day customers.

Another important finding was the relationship between product pricing and customer demand. Lower-priced products were sold in higher quantities, while higher-priced products showed comparatively lower sales volume, demonstrating how pricing influences customer purchasing behavior.

Overall, this project demonstrates how data analytics and business intelligence tools can be effectively used to transform raw transactional data into meaningful business insights. By combining Python-based analysis with Power BI dashboard development, the project provides a comprehensive understanding of coffee shop sales performance and customer behavior. The insights generated from this analysis can help coffee shop businesses optimize operations, improve product strategies, enhance customer satisfaction, and make informed data-driven business decisions.

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CHAPTER 1: INTRODUCTION

1.1 Background

With the rapid growth of the retail and food service industries, coffee shops generate large volumes of transactional data every day. This data contains valuable information related to customer purchasing behavior, sales performance, product demand, pricing, and store operations. Analyzing such data is essential for businesses to understand market trends, improve operational efficiency, optimize product offerings, and enhance customer satisfaction..

1.2 Objectives

- To analyze total sales and revenue performance using the Coffee Shop Sales dataset
- To compare performance across different store locations, product categories, and product types
- To identify top-performing products and peak sales hours
- To study customer purchasing behavior during weekdays and weekends
- To analyze monthly sales trends and business growth over time
- To develop an interactive Power BI dashboard for visualizing business performance and insights
- To generate actionable business insights using data analytics and visualization techniques

1.3 Purpose

The primary purpose of this project is to extract meaningful insights from coffee shop transactional data and understand how different business factors influence sales performance and customer purchasing behavior. The project focuses on analyzing the impact of variables such as product category, product type, product pricing, store location, purchase timing, peak sales hours, and customer behavior during weekdays and weekends.

Another important objective of the project is to demonstrate how data analytics and business intelligence tools such as Python and Power BI can be effectively used to support business decision-making, improve operational efficiency, and enhance revenue generation. Through detailed analysis and visualization, the project helps identify valuable business patterns and trends that can support strategic planning and performance improvement.

1.4 Scope

This project focuses on analyzing coffee shop transactional sales data using Python, SQL, Excel, and Power BI. The scope of the project includes data collection, preprocessing, data cleaning, transformation, feature engineering, exploratory data analysis (EDA), trend analysis, relationship analysis, visualization, and dashboard development using Power BI.

The analysis covers several important business metrics, including total sales and revenue, product demand, category performance, store-wise sales performance, customer purchasing behavior, monthly and hourly sales trends, and weekday versus weekend sales patterns. By analyzing these factors, the project aims to help coffee shop businesses optimize operations, improve marketing strategies, and make effective data-driven business decisions.

1.5 Data Sources

The project uses a Coffee Shop Sales Dataset stored in Excel format. The dataset contains transactional sales data collected from multiple coffee shop locations and includes detailed information related to customer purchases and sales activities.

The dataset consists of various fields such as Transaction ID, Transaction Date, Transaction Time, Transaction Quantity, Store ID, Store Location, Product ID, Unit Price, Product Category, Product Type, Product Detail, Total Sales, Month, Weekday, Hour, and Weekend/Weekday indicators. These attributes provide comprehensive information required for analyzing sales trends, customer behavior, product performance, and store-wise business performance.

1.6 Problem Definition

The main problem addressed in this project is to identify the key factors influencing coffee shop sales and revenue generation.

The project aims to understand:

- How store location impacts business performance
- Which product categories and product types contribute most to revenue
- How product pricing affects customer purchasing behavior

- The impact of weekdays, weekends, and peak business hours on sales
- Customer purchasing trends and demand patterns over time

By analyzing these factors using data analytics and visualization techniques, the project seeks to generate valuable insights that can help improve business performance, customer satisfaction, and strategic decision-making in the coffee shop industry.

CHAPTER 2: TOOLS AND TECHNOLOGY USED

The successful completion of this Coffee Shop Sales Analysis project was made possible through the effective use of various tools and technologies. Each stage of the project—from data loading and preprocessing to analysis, visualization, and dashboard development—required specific tools that ensured accuracy, efficiency, and clarity in generating meaningful business insights.

1. Programming Languages and Libraries

- **Python**

Python served as the core programming language for this project due to its simplicity, flexibility, and powerful ecosystem for data analysis and visualization. It was used for:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature engineering
- Trend analysis
- Visualization of coffee shop sales trends
- Business insight generation

Python provided an efficient environment for handling large transactional datasets and performing analytical operations.

- **Pandas**

Pandas played a major role in handling and analyzing the Coffee Shop Sales dataset.

It was used to:

- Load the dataset using `read_excel()`
 - Perform data cleaning and preprocessing
 - Convert transaction date and transaction time columns into proper datetime format
 - Handle missing values using functions like `isnull()` and filtering techniques
 - Remove duplicate records and ensure data consistency
 - Create a new column called Total Sales using:

$$\text{Transaction Quantity} \times \text{Unit Price}$$

- Extract additional features such as Month, Day, Weekday, Hour, and Weekend indicators
- Perform data aggregation using `groupby()`
- Analyze store-wise, category-wise, and product-level performance
- Sort and filter data using functions like `sort_values()` and `nlargest()`

Pandas significantly improved the efficiency of data manipulation and analysis throughout the project.

- **NumPy**

NumPy was used for numerical computations and efficient handling of large datasets. It supported:

- Mathematical operations on sales and quantity data
- Numerical transformations and calculations
- Statistical operations such as averages, totals, and distributions
- Efficient array-based computations for analytical processing

NumPy enhanced the performance and speed of data operations within the project.

- **Matplotlib**

Matplotlib was used for creating fundamental data visualizations such as:

- Bar charts for store-wise and category-wise sales comparison
- Line charts for daily and monthly sales trends
- Pie charts for category contribution analysis
- Basic plots for understanding sales distributions

It helped in presenting insights in a clear, structured, and understandable format..

- **Seaborn**

Seaborn was used to create advanced and visually appealing statistical graphics. It was used for:

- Bar plots for store, product, and category analysis
- Line plots for hourly sales trend analysis
- Heatmaps to analyze weekday vs hourly sales performance
- Scatter plots to study the relationship between unit price and quantity sold
- Enhanced visualizations with better aesthetics and interpretability

Seaborn improved the visual quality and analytical representation of the dataset

2. Business Intelligence and Dashboard Tool

- **Power BI**

Power BI was used to develop an interactive and professional business intelligence dashboard for the Coffee Shop Sales Analysis project. It enabled effective visualization and presentation of key business insights.

Power BI was used to:

- Create KPI cards for:
 - Total Sales
 - Total Orders
 - Total Quantity Sold
 - Average Sales
- Develop interactive charts and visualizations
- Analyze:
 - Monthly sales trends
 - Store-wise performance
 - Product category contribution
 - Top-selling products
 - Hourly sales performance
 - Weekday vs weekend sales behavior
- Add interactive slicers and filters for dynamic analysis
- Build a single-page dashboard for comprehensive business reporting
- Improve business understanding through visual storytelling

Power BI helped transform analytical results into an interactive dashboard that supports data-driven decision-making.

3. Development Environment

- **Jupyter Notebook**

Jupyter Notebook was used as the primary development environment for performing Python-based analysis. It allowed:

- Writing and executing Python code step-by-step
- Visualizing outputs instantly

- Documenting analysis using Markdown cells
- Combining code, charts, explanations, and outputs in a single interface

This made the entire analytical workflow more organized, interactive, and easier to understand.

- **Microsoft Excel**

Excel was used during the initial stage of the project for:

- Previewing the dataset
- Understanding column structures and data types
- Performing quick validation before analysis in Python
- Verifying sales records and transaction details
- Basic data inspection and formatting

Excel helped in the preliminary understanding and verification of the dataset before advanced analysis.

4. Analytical Techniques Used

The following analytical techniques were applied during the project:

- Data Cleaning and Preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Data Aggregation and Grouping
- Trend Analysis
- Relationship Analysis
- Time-Based Sales Analysis
- Customer Behavior Analysis
- Dashboard Visualization and Reporting

These techniques helped in extracting meaningful insights from raw coffee shop transactional data.

The combination of Python, Pandas, NumPy, Matplotlib, Seaborn, Power BI, Jupyter Notebook, and Excel played a crucial role in successfully completing the Coffee Shop Sales Analysis project. These tools enabled efficient data preprocessing, advanced analysis, interactive visualization, and dashboard development, ultimately helping to transform raw transactional data into valuable business insights and actionable decisions.

CHAPTER 3: DATA COLLECTION AND ANALYSIS

3.1 Data Collection

The dataset used for this project was collected in Excel format and contains detailed information related to coffee shop sales transactions.

The dataset includes fields such as:

- Transaction ID
- Transaction Date
- Transaction Time
- Transaction Quantity
- Store Location
- Product Category
- Product Type
- Product Detail
- Product ID
- Unit Price

The dataset represents daily transactional sales data from multiple coffee shop branches and provides valuable information for analyzing customer purchasing behavior, product demand, store performance, and overall sales trends.

The collected data served as the foundation for performing data preprocessing, exploratory data analysis, visualization, and dashboard development using Python and Power BI.

3.2 Data Preprocessing

Data preprocessing was an important step in ensuring the quality, consistency, and reliability of the dataset before analysis. The following preprocessing techniques were performed:

- Checked for missing values and verified overall data completeness
- Converted Transaction Date and Transaction Time columns into proper datetime format
- Cleaned inconsistent entries and ensured uniform data types across all columns
- Checked for duplicate records and removed them where necessary
- Verified numerical columns such as Transaction Quantity and Unit Price for accurate calculations

- Created a new column called Total Sales using the formula:
Total Sales = Transaction Quantity × Unit Price
- Standardized column names and formats for easier analysis and reporting
- Validated data consistency across product categories and store locations

These preprocessing steps improved data accuracy and prepared the dataset for detailed analytical operations.

3.3 Feature Engineering

Feature engineering techniques were applied to generate additional variables and improve the analytical capabilities of the dataset. The following features were created:

- Extracted Month from Transaction Date for monthly sales trend analysis
- Extracted Day and Weekday to study weekday and weekend purchasing behavior
- Extracted Hour from Transaction Time to identify peak business hours
- Created a Weekend/Weekday Flag for customer behavior analysis
- Aggregated data at different levels such as:
 - Store Location
 - Product Category
 - Product Type
- Created summary metrics including:
 - Total Revenue
 - Average Order Value
 - Total Quantity Sold
 - Transformed data into grouped and summarized formats for easier comparison across stores and product categories

Feature engineering helped in uncovering deeper business insights and enabled detailed time-based and category-wise analysis.

3.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to identify trends, patterns, relationships, and business insights within the coffee shop sales data. Various statistical methods and visualizations were used to analyze sales performance, customer purchasing behavior, and product demand.

Key Findings from the Analysis

- Among all store locations, Hell's Kitchen generated the highest revenue, followed by Astoria and Lower Manhattan, indicating balanced store performance and strong customer engagement across locations.
- Coffee was identified as the highest-selling product category, followed by Tea and Bakery, showing strong customer preference for beverage products.
- Morning hours between 8 AM and 10 AM showed the highest sales activity, with 10 AM recording the maximum revenue, indicating strong demand during breakfast and office hours.
- Weekday sales were significantly higher than weekend sales, suggesting that regular working-day customers contributed more consistently to business revenue.
- Monthly sales trends showed steady business growth, with the highest sales recorded during May and June, indicating increasing customer demand and improving business performance over time.

Major Insight

One of the most important insights identified during the analysis was the strong relationship between Product Price and Quantity Sold.

The analysis showed that:

- Lower-priced products were sold in higher quantities
- Higher-priced products were purchased less frequently

This indicates that product pricing has a direct influence on customer purchasing behavior and overall sales volume.

The finding highlights the importance of maintaining balanced pricing strategies to maximize both:

- Sales volume
- Business profitability

Understanding this relationship can help coffee shop businesses optimize pricing, improve product strategies, and make better data-driven business decisions.

CHAPTER 4: SYSTEM REQUIREMENTS AND ANALYSIS

4.1 System Requirements Specification

This project requires a well-structured analytical system to efficiently process, analyze, visualize, and interpret coffee shop sales data. The system is designed to support smooth execution of data analytics and business intelligence tasks while ensuring accuracy, reliability, scalability, and effective presentation of business insights.

The project integrates Python-based data analysis techniques with Power BI dashboard development to transform raw transactional data into meaningful and actionable insights.

4.1.1 Functional Requirements

The functional requirements define the major operations and analytical tasks that the system must perform to achieve the objectives of the project. The system should be capable of loading the Coffee Shop Sales dataset from an Excel file, performing data cleaning and preprocessing operations, handling missing values and duplicate records, and converting transaction date and time columns into proper datetime formats.

The system should also calculate Total Sales using $\text{Transaction Quantity} \times \text{Unit Price}$ and create additional analytical features such as Month, Day, Weekday, Hour, and Weekend/Weekday indicators. These features support detailed analysis of sales trends and customer purchasing behavior.

In addition, the system should perform sales analysis across multiple dimensions including:

- Store locations
- Product categories
- Product types
- Time periods

The system should further identify top-performing products and peak sales hours, analyze customer purchasing behavior during weekdays and weekends, generate statistical summaries and analytical reports, create meaningful visualizations, and develop an interactive Power BI dashboard. It should also support dynamic filtering and

visualization of sales data while generating insights that assist in business decision-making.

4.1.2 Non-Functional Requirements

The non-functional requirements define the quality standards, usability, and performance expectations of the system. The system should provide efficient and fast data processing along with accurate and reliable analytical results. It should also generate clear, interactive, and easy-to-understand visualizations that improve the communication of business insights.

The system must support scalability for handling large transactional datasets and ensure consistent performance during analysis and visualization tasks. Additionally, the dashboard and reporting system should be user-friendly, easy to maintain, and capable of future enhancements. High readability and structured presentation of insights are also important requirements to ensure effectiveness in real-world business analytics applications.

4.2 Hardware and Software Requirements

To successfully implement the Coffee Shop Sales Analysis project, specific hardware and software resources are required.

- **Hardware Requirements**
 - System: Laptop/Desktop
 - Processor: Intel i3 or above
 - RAM: Minimum 4 GB (8 GB recommended)
 - Storage: Minimum 10 GB free space
- **Software Requirements**
 - Operating System: Windows / Linux / macOS
 - Programming Language: Python
 - Development Environment: Jupyter Notebook
 - Business Intelligence Tool: Power BI
 - Spreadsheet Tool: Microsoft Excel
- **Python Libraries Used**
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn

- **Dataset Format**

Excel File (.xlsx)

These tools and technologies support data preprocessing, analysis, visualization, and dashboard development throughout the project.

4.3 System Overview

The system is designed to analyze coffee shop sales data through a structured analytical workflow that combines data preprocessing, exploratory data analysis, visualization, and business intelligence dashboard development. The workflow begins with loading the transactional dataset, followed by cleaning and preparing the data for analysis. Additional analytical features are then created to support detailed sales and customer behavior analysis.

The overall workflow of the system includes:

1. Importing the Coffee Shop Sales dataset
2. Cleaning and preprocessing the data
3. Handling missing values and duplicate records
4. Converting date and time columns into proper datetime format
5. Creating analytical features such as Total Sales, Month, Weekday, Hour, and Weekend/Weekday indicators
6. Performing Exploratory Data Analysis (EDA)
7. Analyzing sales trends across stores, product categories, product types, and time periods
8. Creating visualizations such as bar charts, line charts, pie charts, heatmaps, and scatter plots
9. Developing an interactive Power BI dashboard
10. Generating business insights and conclusions

The system transforms raw coffee shop transactional data into meaningful business intelligence that helps organizations improve sales performance, understand customer purchasing behavior, optimize product strategies, monitor store performance, support data-driven decision-making, and enhance overall customer satisfaction.

By integrating Python-based analytics with Power BI dashboard visualization, the system provides a comprehensive and effective solution for coffee shop sales analysis.

CHAPTER 5: IMPLEMENTATION

5.1 System Design

The Coffee Shop Sales Analysis system is designed to process, analyze, visualize, and interpret transactional sales data in a structured and efficient manner. The system follows a step-by-step analytical workflow that transforms raw sales data into meaningful business insights and interactive visual reports.

The overall system consists of three major components: input, processing, and output. The input component includes the Coffee Shop Sales dataset stored in Excel format, containing transactional details such as transaction date and time, store location, product category, product type, product detail, transaction quantity, and unit price.

The processing stage involves multiple analytical operations including data cleaning, preprocessing, feature engineering, exploratory data analysis (EDA), statistical analysis, visualization, and dashboard development. During this stage, the data is processed using Python libraries such as Pandas, NumPy, Matplotlib, and Seaborn, along with Power BI for creating interactive dashboards and reports.

The output component generates valuable business insights such as sales performance analysis, customer purchasing behavior analysis, product performance reports, and interactive visualizations. The final output helps coffee shop businesses understand trends, optimize operations, and improve overall business performance through data-driven decision-making.

5.2 Workflow

The project follows a structured workflow to ensure accurate analysis and meaningful insight generation. Initially, the Coffee Shop Sales dataset was imported from an Excel file using Pandas and loaded into the Python environment for analysis.

The next stage involved data cleaning and preprocessing, where missing values and inconsistencies were checked, duplicate records were removed, and appropriate data types were assigned to numerical and datetime columns. Column formats were also standardized to improve consistency and simplify further analysis.

Feature engineering was then performed to create additional analytical variables such as Total Sales, Month, Day, Weekday, Hour, and Weekend Flag. These features enabled detailed time-based analysis and improved understanding of customer purchasing behavior.

After preprocessing, Exploratory Data Analysis (EDA) was conducted to identify sales trends, customer behavior patterns, product performance, store-wise revenue distribution, peak sales hours, and differences between weekday and weekend sales. This analysis helped uncover meaningful relationships and business patterns within the dataset.

Visualization techniques were applied using Matplotlib, Seaborn, and Power BI to represent the data through bar charts, line charts, pie charts, heatmaps, scatter plots, and interactive dashboards. These visualizations improved the interpretation and presentation of analytical findings.

Finally, the insights generated from the analysis were interpreted to draw conclusions and provide business recommendations. The project identified key factors affecting sales and customer behavior, helping support data-driven business decisions and performance improvement strategies.

5.3 Modules

The project is divided into multiple functional modules to ensure systematic processing and analysis of the sales data.

- **Data Processing Module**

The Data Preprocessing Module is responsible for loading the dataset, cleaning missing and duplicate records, converting date and time columns, and verifying data consistency and accuracy. This module ensures that the dataset is reliable and ready for analysis.

- **Feature Engineering Module**

The Feature Engineering Module focuses on creating new analytical features such as Month, Weekday, Hour, Weekend indicators, and Total Sales. It also prepares grouped and summarized data required for detailed analysis. This module improves the analytical depth of the project and supports advanced business insights.

- **Analysis Module**

The Analysis Module performs exploratory data analysis, sales trend analysis, product performance evaluation, customer purchasing behavior analysis, store-wise revenue analysis, and time-based sales analysis. This module helps identify important trends, relationships, and business patterns from the dataset.

- **Visualization Module**

The Visualization Module creates graphical representations of the data using Matplotlib, Seaborn, and Power BI. Various charts and dashboards are developed to simplify complex information and improve communication of insights.

- **Reporting and Insight Generation Module**

The Reporting and Insight Generation Module is responsible for interpreting analytical results, generating business insights, preparing reports and conclusions, and developing dashboards for presentation and decision-making. This module helps transform analytical findings into practical recommendations that can support business growth and operational improvement. findings into actionable business knowledge.

5.4 Project Screenshot

1. Microsoft PowerBI Dashboard



Figure 1 : Dashboard

2. Importing required libraries and loading the dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load dataset
df = pd.read_excel("Coffee Shop Sales.xlsx")
df.head()
```

	transaction_id	transaction_date	transaction_time	transaction_qty	store_id	store_location	product_id	unit_price	product_category	product_type	product_detail
0	1	2023-01-01	07:06:11	2	5	Lower Manhattan	32	3.0	Coffee	Gourmet brewed coffee	Ethiopia Rg
1	2	2023-01-01	07:08:56	2	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
2	3	2023-01-01	07:14:04	2	5	Lower Manhattan	59	4.5	Drinking Chocolate	Hot chocolate	Dark chocolate Lg
3	4	2023-01-01	07:20:24	1	5	Lower Manhattan	22	2.0	Coffee	Drip coffee	Our Old Time Diner Blend Sm
4	5	2023-01-01	07:22:41	2	5	Lower Manhattan	57	3.1	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg

Figure 2: Dataset Overview

3. Initial Data Exploration

```
print(df.info())  
print(df.describe())
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 149116 entries, 0 to 149115  
Data columns (total 11 columns):  
#   Column                Non-Null Count  Dtype    
---  ---                    -  
0   transaction_id         149116 non-null int64    
1   transaction_date       149116 non-null datetime64[ns]  
2   transaction_time       149116 non-null object   
3   transaction_qty        149116 non-null int64    
4   store_id              149116 non-null int64    
5   store_location        149116 non-null object   
6   product_id            149116 non-null int64    
7   unit_price            149116 non-null float64   
8   product_category      149116 non-null object   
9   product_type          149116 non-null object   
10  product_detail         149116 non-null object   
dtypes: datetime64[ns](1), float64(1), int64(4), object(5)  
memory usage: 12.5+ MB  
None
```

	transaction_id	transaction_date	transaction_qty \
count	149116.000000	149116	149116.000000
mean	74737.371872	2023-04-15 11:50:32.173609984	1.438276
min	1.000000	2023-01-01 00:00:00	1.000000
25%	37335.750000	2023-03-06 00:00:00	1.000000
50%	74727.500000	2023-04-24 00:00:00	1.000000
75%	112094.250000	2023-05-30 00:00:00	2.000000
max	149456.000000	2023-06-30 00:00:00	8.000000
std	43153.600016	NaN	0.542509

	store_id	product_id	unit_price
count	149116.000000	149116.000000	149116.000000
mean	5.342063	47.918607	3.382219
min	3.000000	1.000000	0.800000
25%	3.000000	33.000000	2.500000
50%	5.000000	47.000000	3.000000
75%	8.000000	60.000000	3.750000
max	8.000000	87.000000	45.000000
std	2.074241	17.930020	2.658723

Figure 3: Data Exploration

4. Data Cleaning and Preprocessing

- Date and Time Conversion
- Check Missing Values
- Duplicate Removal
- Data Type Correction

```
# Convert date & time
df['transaction_date'] = pd.to_datetime(df['transaction_date'])
df['transaction_time'] = pd.to_datetime(df['transaction_time'], format='%H:%M:%S').dt.time

# Check missing values
print(df.isnull().sum())

# Drop duplicates
df = df.drop_duplicates()

# Fix data types
df['transaction_qty'] = df['transaction_qty'].astype(int)
df['unit_price'] = df['unit_price'].astype(float)

# Create total sales column
df['total_sales'] = df['transaction_qty'] * df['unit_price']

transaction_id    0
transaction_date  0
transaction_time  0
transaction_qty   0
store_id          0
store_location    0
product_id        0
unit_price        0
product_category  0
product_type      0
product_detail    0
dtype: int64
```

Figure 3: Data cleaning

```

# Check missing values
print(df.isnull().sum())
print(df.duplicated().sum())

transaction_id      0
transaction_date    0
transaction_time     0
transaction_qty     0
store_id            0
store_location      0
product_id          0
unit_price          0
product_category    0
product_type        0
product_detail      0
total_sales         0
year               0
month              0
day               0
weekday            0
hour               0
is_weekend         0
dtype: int64
0

# Convert date & time
df['transaction_date'] = pd.to_datetime(df['transaction_date'])
df['transaction_time'] = pd.to_datetime(df['transaction_time'], format='%H:%M:%S').dt.time

# Drop duplicates
df = df.drop_duplicates()

# Fix data types
df['transaction_qty'] = df['transaction_qty'].astype(int)
df['unit_price'] = df['unit_price'].astype(float)

df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 149116 entries, 0 to 149115
Data columns (total 11 columns):
#   Column                Non-Null Count  Dtype
---  -
0   transaction_id         149116 non-null  int64
1   transaction_date       149116 non-null  datetime64[ns]
2   transaction_time       149116 non-null  object
3   transaction_qty        149116 non-null  int64
4   store_id              149116 non-null  int64
5   store_location         149116 non-null  object
6   product_id            149116 non-null  int64
7   unit_price            149116 non-null  float64
8   product_category      149116 non-null  object
9   product_type          149116 non-null  object
10  product_detail         149116 non-null  object
dtypes: datetime64[ns](1), float64(1), int64(4), object(5)
memory usage: 12.5+ MB

```

Figure 4: Data preprocessing

5. Store Wise Sales Analysis

```
store_sales = df.groupby('store_location')['total_sales'].sum().sort_values()  
store_sales
```

```
store_location  
Lower Manhattan    230057.25  
Astoria            232243.91  
Hell's Kitchen     236511.17  
Name: total_sales, dtype: float64
```

```
store_sales.plot(kind='barh')  
plt.title("Revenue by Store Location")  
plt.show()
```

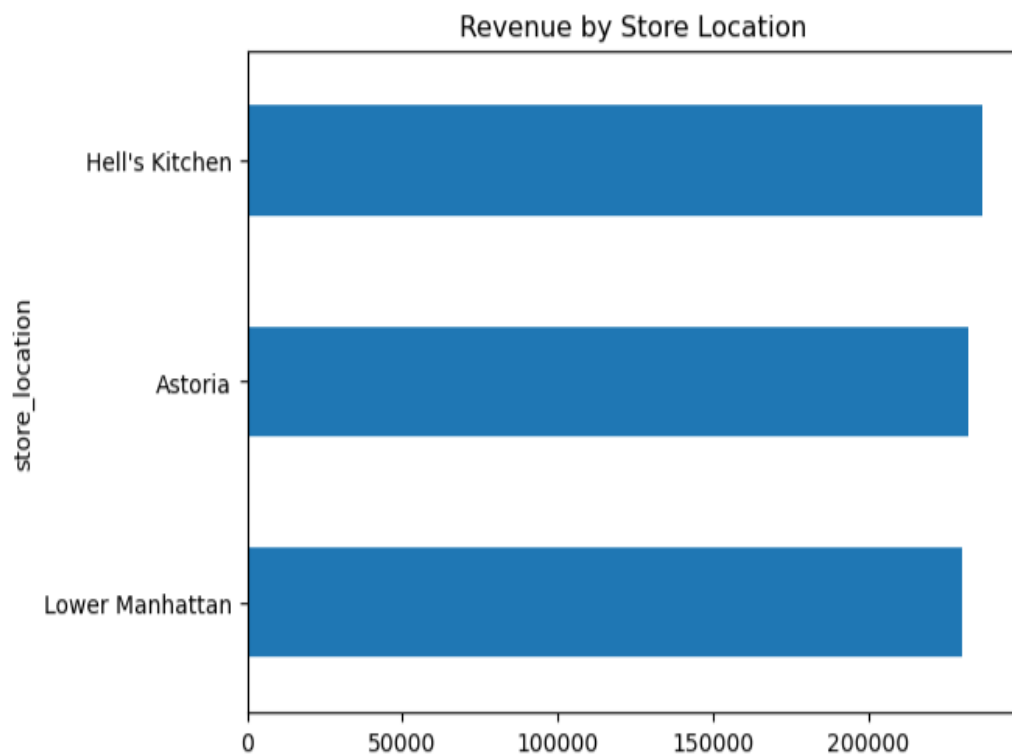


Figure 5: Store wise sales

6. Daily Sales Trend Analysis

```
daily_sales = df.groupby('transaction_date')['total_sales'].sum()  
daily_sales
```

```
transaction_date  
2023-01-01    2508.20  
2023-01-02    2403.35  
2023-01-03    2565.00  
2023-01-04    2220.10  
2023-01-05    2418.85  
...  
2023-06-26    5875.90  
2023-06-27    5975.65  
2023-06-28    4728.90  
2023-06-29    4450.75  
2023-06-30    5481.32  
Name: total_sales, Length: 181, dtype: float64
```

```
daily_sales.plot()  
plt.title("Daily Sales Trend")  
plt.show()
```

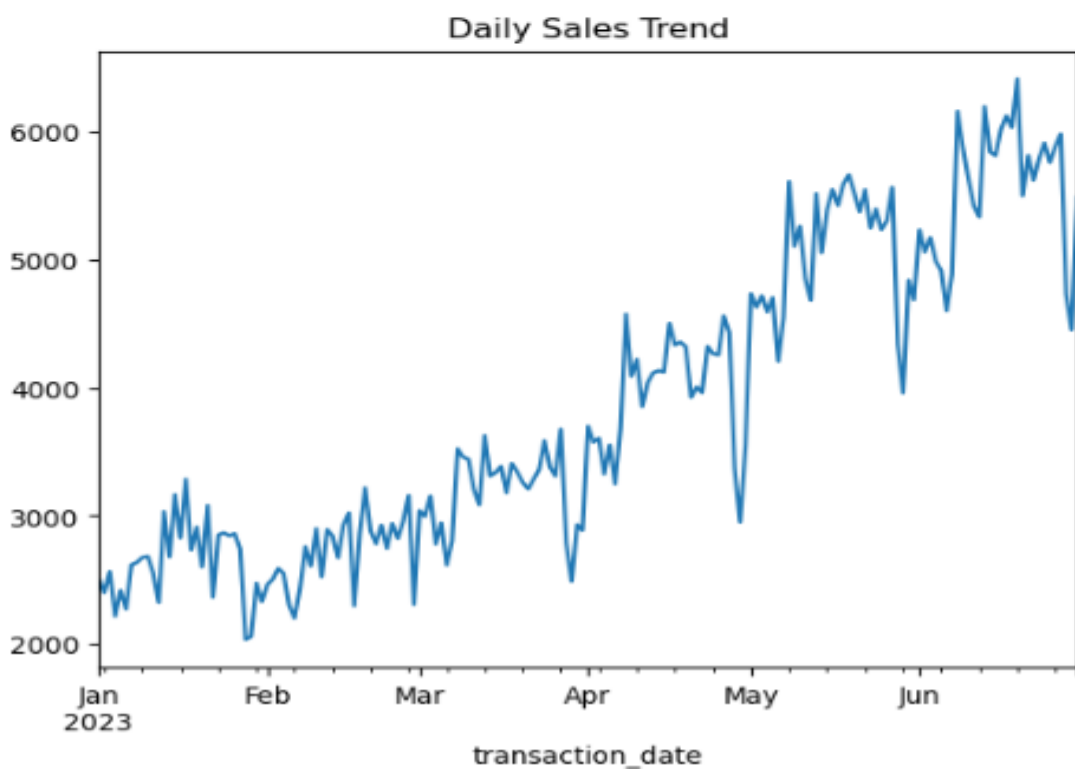


Figure 6: Daily sales

7. Monthly Sales Analysis

```
monthly_sales = df.groupby('month')['total_sales'].sum()
monthly_sales
```

```
month
1      81677.74
2      76145.19
3      98834.68
4     118941.08
5     156727.76
6     166485.88
Name: total_sales, dtype: float64
```

```
sns.barplot(x=monthly_sales.index, y=monthly_sales.values)
plt.title("Monthly Sales")
plt.show()
```

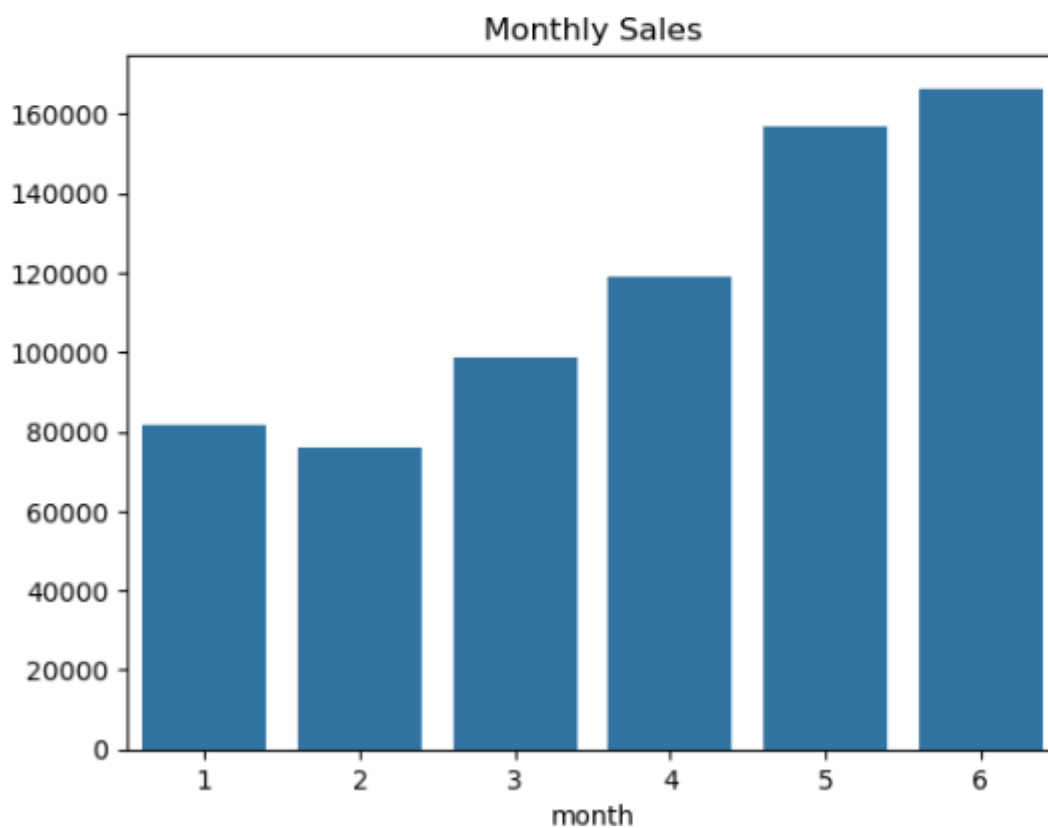


Figure 7: Monthly sales

8. Product Category Analysis

```
category_sales = df.groupby('product_category')['total_sales'].sum().nlargest(5)  
category_sales
```

```
product_category  
Coffee          269952.45  
Tea             196405.95  
Bakery          82315.64  
Drinking Chocolate  72416.00  
Coffee beans    40085.25  
Name: total_sales, dtype: float64
```

```
category_sales.plot(kind='pie', autopct='%1.1f%%')  
plt.title("Sales by Category")  
plt.show()
```

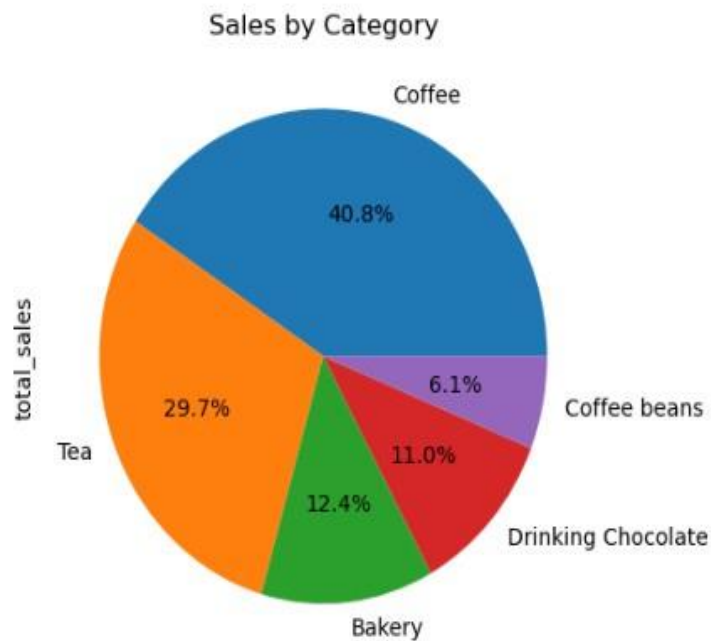


Figure 8: Product category

9. Top 10 Products by Total Sales

```
top_products = df.groupby('product_detail')['total_sales'].sum().nlargest(10)
top_products
```

```
product_detail
Sustainably Grown Organic Lg    21151.75
Dark chocolate Lg              21006.00
Latte Rg                       19112.25
Cappuccino Lg                  17641.75
Morning Sunrise Chai Lg        17384.00
Latte                          17257.50
Jamaican Coffee River Lg       16481.25
Sustainably Grown Organic Rg   16233.75
Cappuccino                     15997.50
Brazilian Lg                   15109.50
Name: total_sales, dtype: float64
```

```
sns.barplot(x=top_products.values, y=top_products.index)
plt.title("Top 10 Products")
plt.show()
```

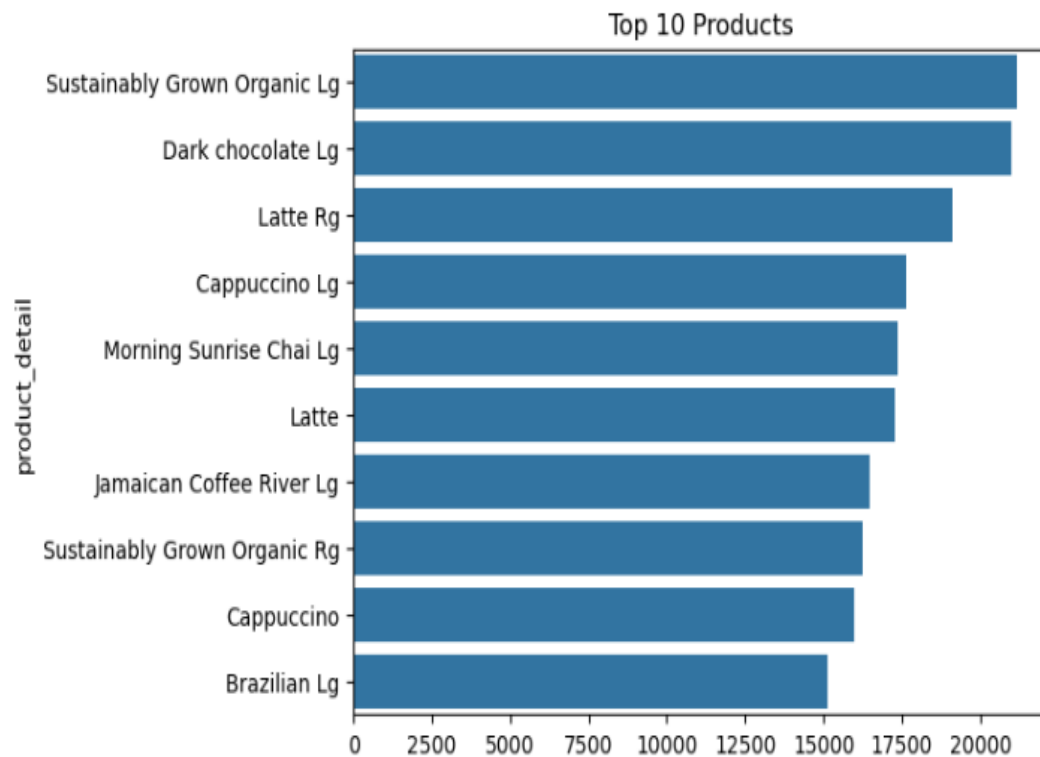


Figure 9: Top 10 products

10. Hourly Sales Analysis

```
hourly_sales = df.groupby('hour')['total_sales'].sum()
hourly_sales
```

```
hour
6    21900.27
7    63526.47
8    82699.87
9    85169.53
10   88673.39
11   46319.14
12   40192.79
13   40367.45
14   41304.74
15   41733.10
16   41122.75
17   40134.31
18   34286.20
19   28446.68
20    2935.64
Name: total_sales, dtype: float64
```

```
sns.lineplot(x=hourly_sales.index, y=hourly_sales.values)
plt.title("Sales by Hour")
plt.show()
```

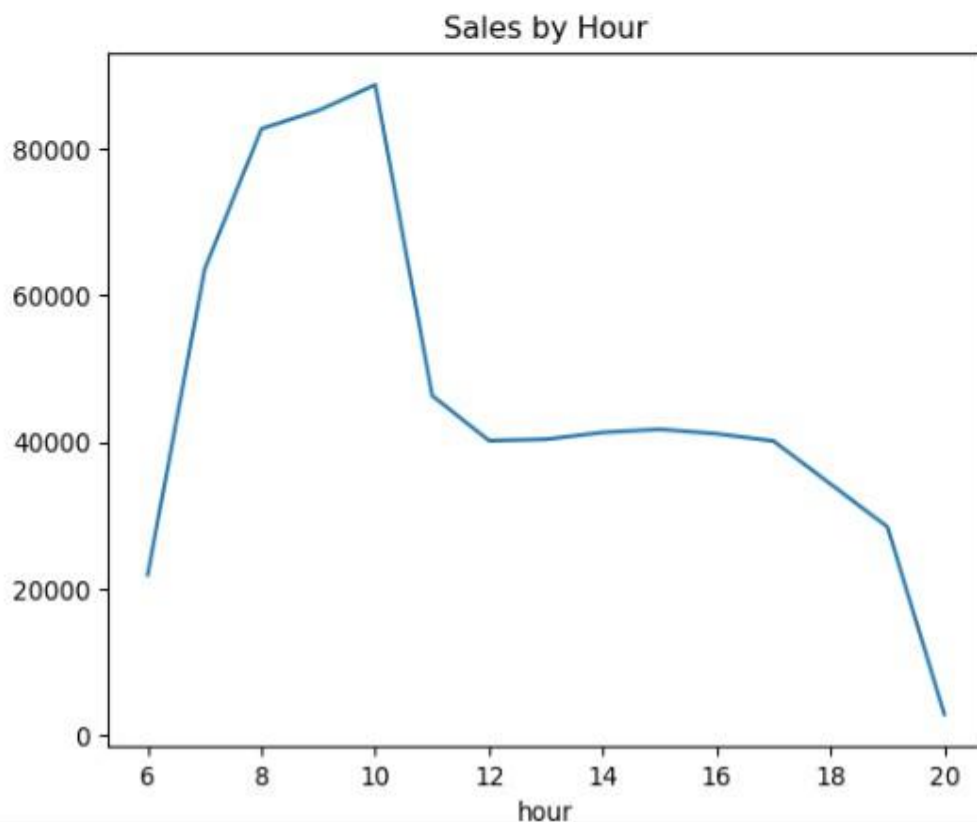


Figure 10: Hourly Sales

11. Weekday vs Weekend Analysis

```
weekend_sales = df.groupby('is_weekend')['total_sales'].sum()
weekend_sales
```

```
is_weekend
False    503587.54
True     195224.79
Name: total_sales, dtype: float64
```

```
sns.barplot(x=weekend_sales.index, y=weekend_sales.values)
plt.title("Weekend vs Weekday Sales")
plt.show()
```

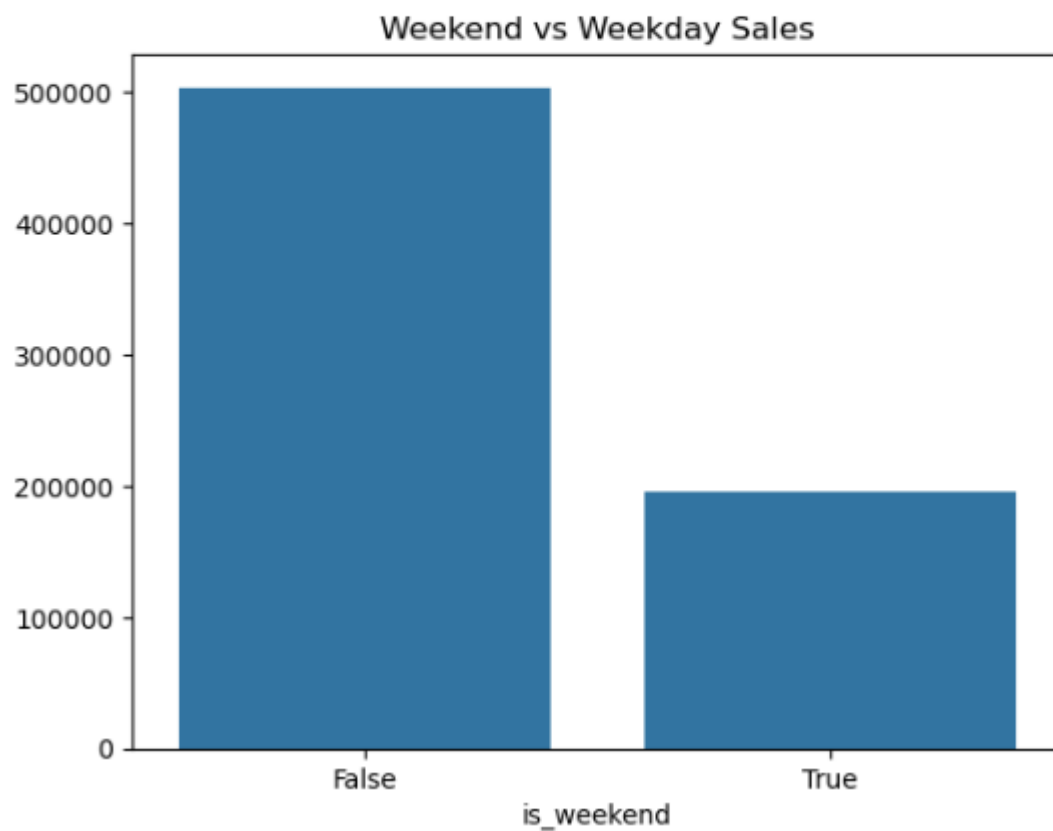


Figure 11: Weekend vs Weekday

12. Weekday Sales

```
weekday_sales = df.groupby('weekday')['total_sales'].sum()  
weekday_sales
```

```
weekday  
Friday      101373.00  
Monday      101677.28  
Saturday     96894.48  
Sunday       98330.31  
Thursday    100767.78  
Tuesday      99455.94  
Wednesday   100313.54  
Name: total_sales, dtype: float64
```

```
sns.barplot(x=weekday_sales.index, y=weekday_sales.values)  
plt.xticks(rotation=45)  
plt.title("Sales by Weekday")  
plt.show()
```

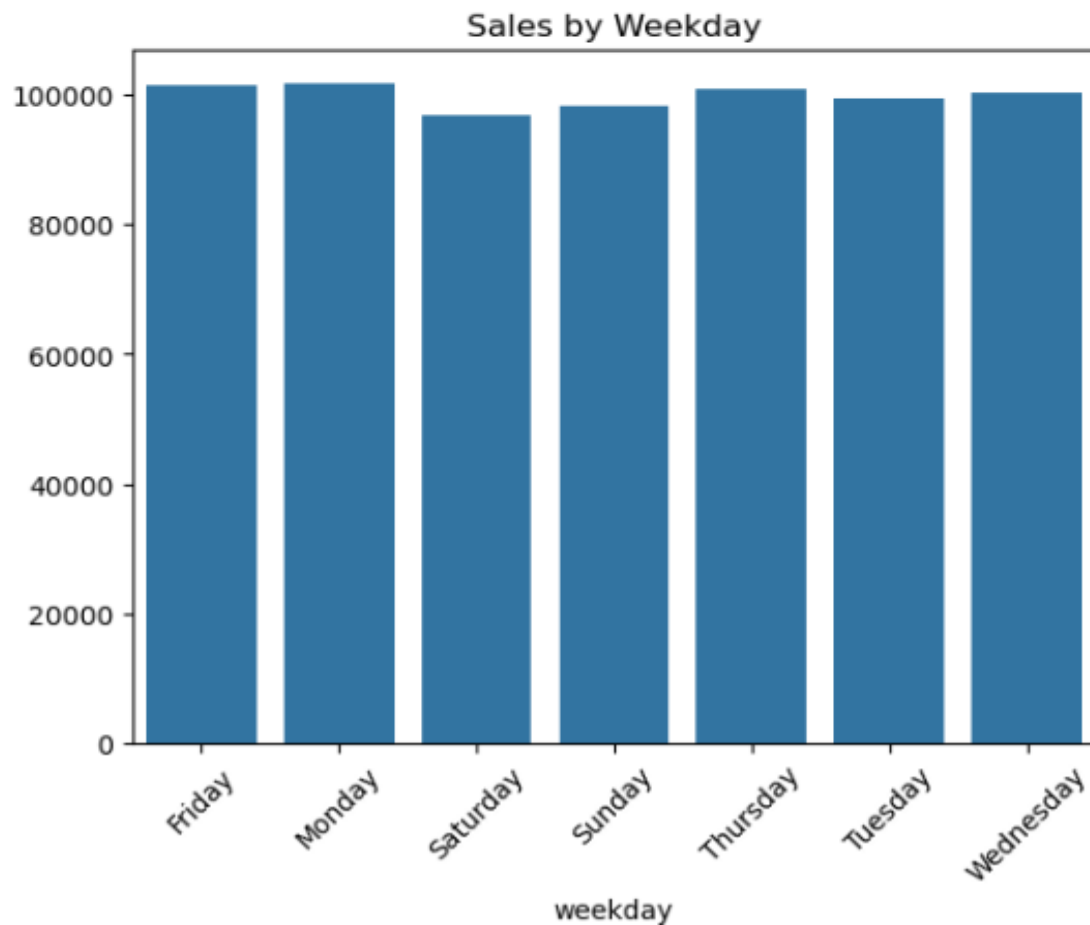


Figure 13: Weekday Sales

13. Quantity Sold by Category

```
avg_order_value = df.groupby('transaction_id')['total_sales'].sum().mean()
print(avg_order_value)
```

```
4.6863671986435265
```

```
qty_category = df.groupby('product_category')['transaction_qty'].sum()
qty_category
```

```
product_category
Bakery          23214
Branded         776
Coffee         89258
Coffee beans    1828
Drinking Chocolate 17457
Flavours       18511
Loose Tea       1218
Packaged Chocolate 487
Tea           69737
Name: transaction_qty, dtype: int64
```

```
qty_category.plot(kind='bar')
plt.title("Quantity Sold by Category")
plt.show()
```

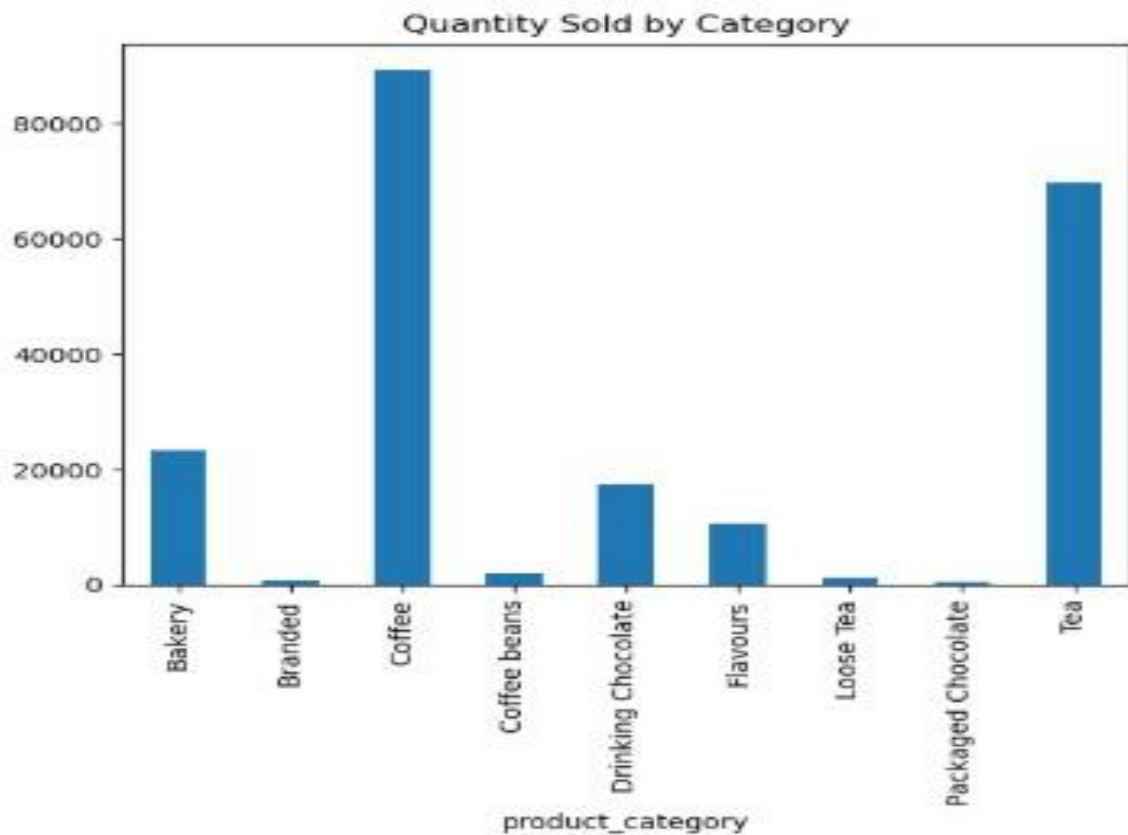


Figure 14: Quantity Sold by Category

14. Transaction Volume by Store

```
store_transactions = df.groupby('store_location')['transaction_id'].count()  
store_transactions
```

```
store_location  
Astoria          50599  
Hell's Kitchen   50735  
Lower Manhattan  47782  
Name: transaction_id, dtype: int64
```

```
sns.barplot(x=store_transactions.index, y=store_transactions.values)  
plt.xticks(rotation=45)  
plt.title("Transactions by Store")  
plt.show()
```

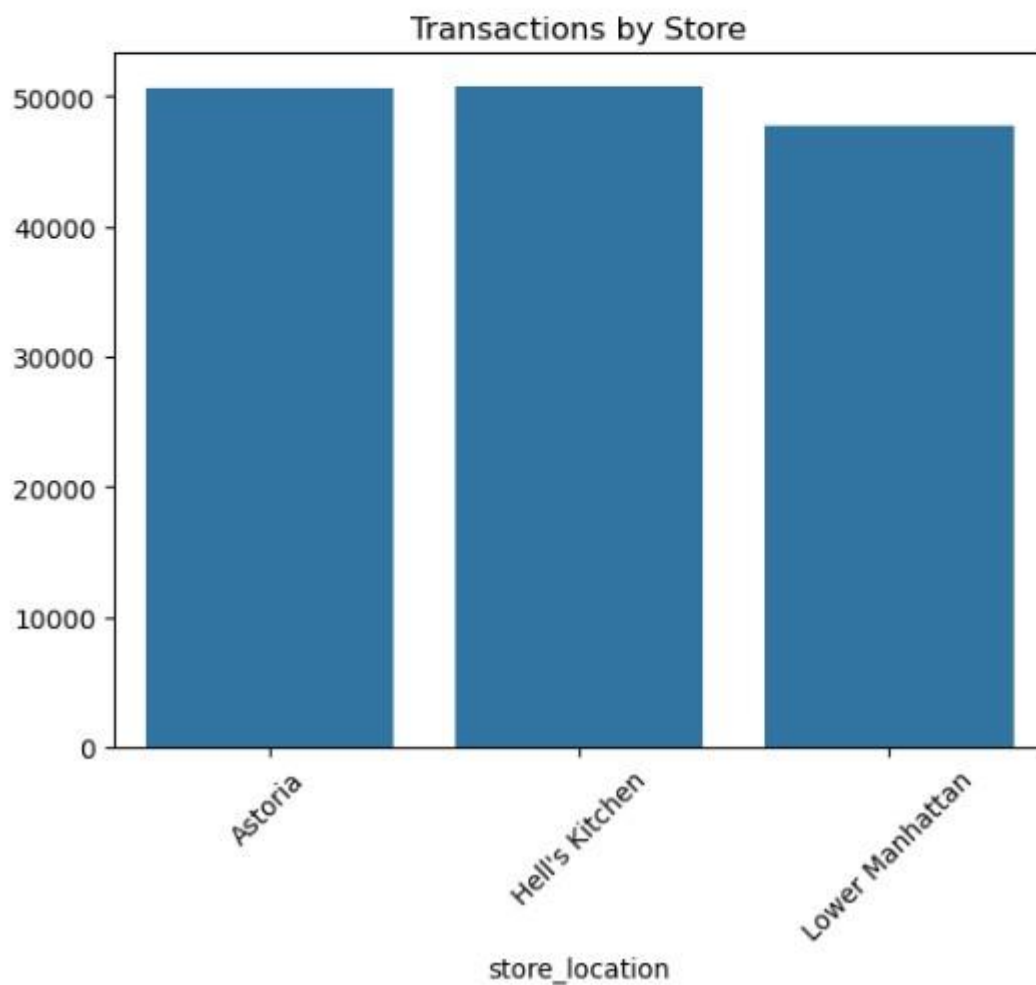


Figure 15: Transaction Volume by Store

15. Sales HeatMap (weekday vs hour)

```
pivot = df.pivot_table(values='total_sales', index='weekday', columns='hour', aggfunc='sum')
pivot
```

hour	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
weekday															
Friday	3385.73	9386.04	11746.65	12750.00	13155.66	6827.84	5685.49	5499.49	5840.33	6096.13	5654.93	5687.14	5139.84	4128.20	387.53
Monday	3098.58	9516.52	11604.69	11898.33	12427.25	7179.09	6139.64	5771.26	6463.31	6159.49	6139.96	5927.35	5006.19	4015.22	330.40
Saturday	2913.85	8775.02	11370.86	11333.85	12736.48	6361.41	5957.75	5549.45	5626.89	5946.50	5825.73	5444.70	4642.40	3970.96	438.63
Sunday	3320.73	8892.09	11615.54	12388.75	12382.75	6118.99	5622.99	5908.18	5765.62	5614.75	5728.46	5788.39	4792.48	4006.71	383.88
Thursday	3464.90	9209.23	11420.49	10963.73	12091.09	6898.55	6073.93	5972.46	6114.61	6087.68	6400.13	6027.33	5234.58	4529.10	279.97
Tuesday	2320.25	8841.71	12950.19	13588.47	13497.68	5877.41	5172.64	5443.46	5621.56	5852.24	5723.78	5609.85	4304.36	3967.94	684.40
Wednesday	3396.23	8905.86	11991.45	12246.40	12382.48	7055.85	5540.35	6223.15	5872.42	5974.31	5649.76	5649.55	5166.35	3828.55	430.83

```
sns.heatmap(pivot, cmap='coolwarm')
plt.title("Heatmap: Sales (weekday vs Hour)")
plt.show()
```

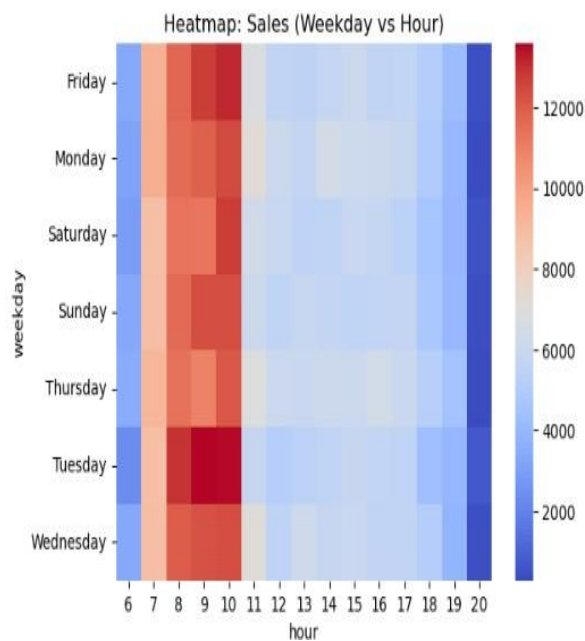


Figure 16: Sales Heatmap

16. Top Products Types

```
type_sales = df.groupby('product_type')['total_sales'].sum().nlargest(10)
type_sales
```

```
product_type
Barista Espresso      91406.20
Brewed Chai tea       77081.95
Hot chocolate         72416.00
Gourmet brewed coffee  70034.60
Brewed Black tea       47932.00
Brewed herbal tea      47539.50
Premium brewed coffee  38781.15
Organic brewed coffee  37746.50
Scone                 36866.12
Drip coffee            31984.00
Name: total_sales, dtype: float64
```

```
sns.barplot(x=type_sales.values, y=type_sales.index)
plt.title("Top Product Types")
plt.show()
```

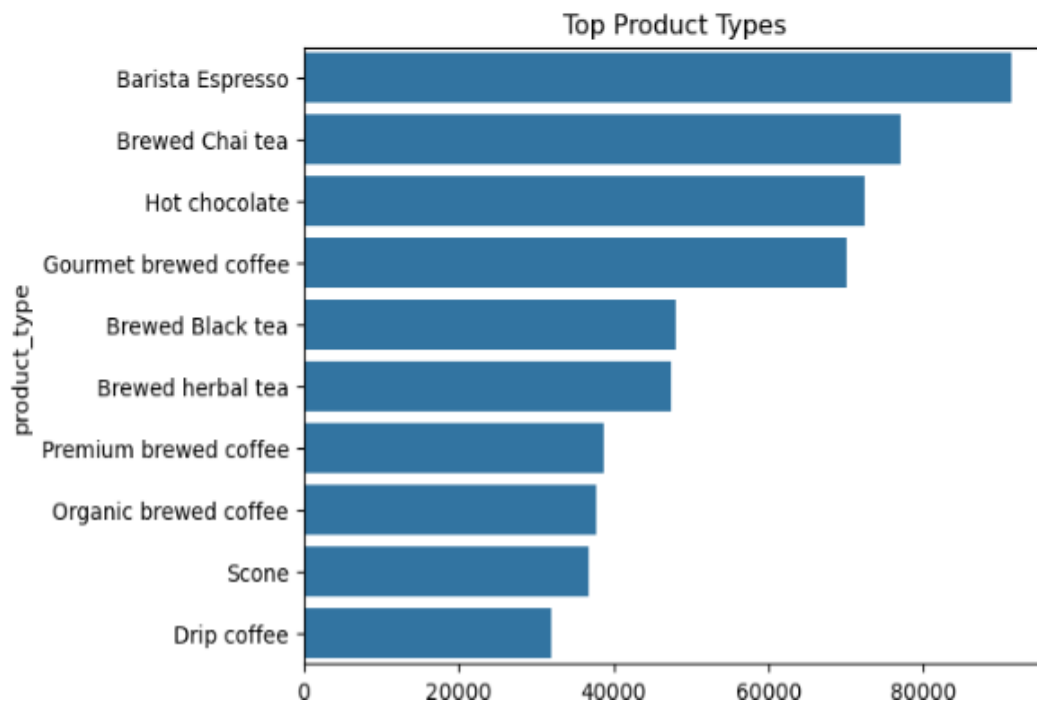


Figure 17: Top Products Types

17. Price vs Quantity Analysis

```
sns.scatterplot(data=df, x='unit_price', y='transaction_qty')  
plt.title("Price vs Quantity")  
plt.show()
```

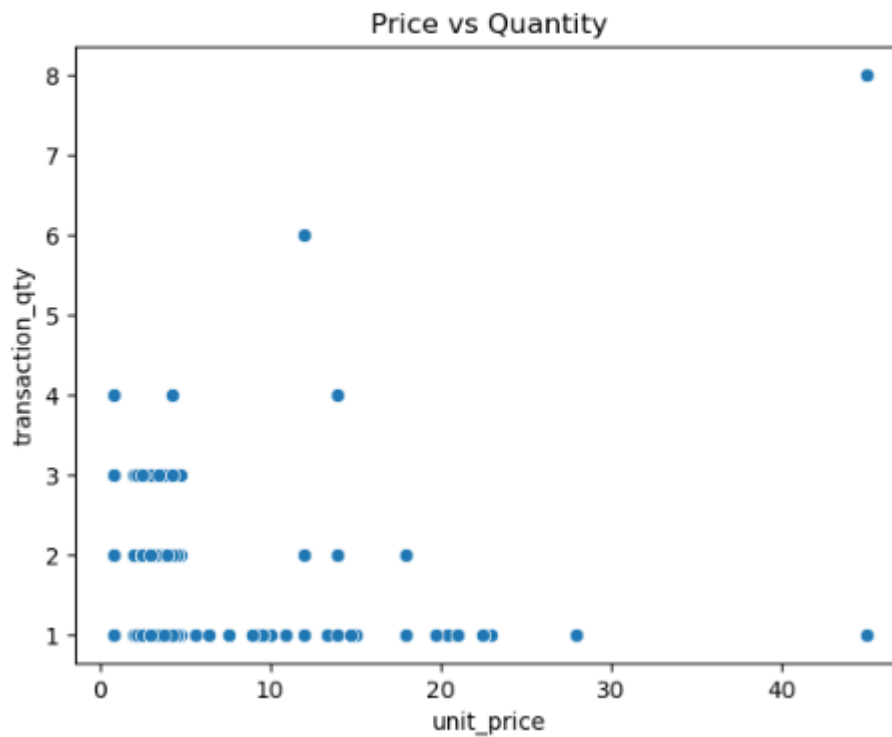


Figure 18: Price vs Quantity Analysis

5.5 REPORT ON THE ANALYSIS

The Coffee Shop Sales Analysis project provided detailed insights into sales performance, customer purchasing behavior, product demand, and store-wise revenue generation. The analysis was performed using Python, data visualization libraries, and Power BI dashboards to identify meaningful business trends and patterns.

Store-wise performance analysis revealed that Hell's Kitchen emerged as the top-performing store with a total revenue of approximately 236,511.17, followed closely by Astoria (232,243.91) and Lower Manhattan (230,057.25). The relatively small difference in revenue among the three stores indicates that all locations performed consistently, with Hell's Kitchen slightly leading in overall sales performance.

Analysis of product categories showed that Coffee dominated total sales with revenue of 269,952.45, followed by Tea (196,405.95) and Bakery (82,315.64). This clearly indicates that beverages are the primary revenue drivers and form the core business segment of the coffee shop.

Quantity-wise analysis also confirmed strong customer preference for beverage products. Coffee recorded the highest quantity sold with 89,250 units, followed by Tea with 69,737 units sold. This suggests that customers frequently purchase coffee and tea products as part of their daily routines.

Product-level analysis identified several top-performing products that contributed significantly to overall revenue. Products such as:

- Sustainably Grown Organic Large (21,151.75)
- Dark Chocolate Large (21,006.00)
- Latte Regular (19,112.25)

generated the highest revenue and showed strong customer demand.

At the same time, certain product categories contributed comparatively lower revenue. Categories such as:

- Packaged Chocolate (4,407.64)
- Flavours (8,408.80)
- Loose Tea (11,213.60)

performed weakly in comparison to beverages. This highlights a noticeable gap between high-performing and low-performing categories and suggests opportunities for improved marketing strategies, product positioning, and promotional activities.

Monthly sales trend analysis showed continuous business growth over time. June recorded the highest sales revenue of 166,485.88, followed by May (156,727.76), while January (81,677.74) and February (76,145.19) recorded the lowest sales. This indicates increasing customer demand and improved business performance during later months.

Hourly sales analysis revealed that 10 AM generated the highest revenue (88,673.39), followed by 9 AM and 8 AM. This clearly shows that morning hours are the busiest period for the coffee shop, likely driven by breakfast customers and office-going consumers. In contrast, sales dropped significantly during evening hours, with 8 PM generating only 2,935.64, indicating lower customer activity later in the day.

Customer purchasing behavior analysis showed that weekday sales (503,587.54) were significantly higher than weekend sales (195,224.79). This suggests that the coffee shop receives stronger customer traffic during regular working days compared to weekends.

Among weekdays, Monday (101,677.28) and Friday (101,373.00) recorded the highest sales, while Tuesday (99,455.94) showed slightly lower performance. Overall, sales remained relatively stable throughout the weekdays with only minor fluctuations.

Store transaction analysis further supported the revenue findings. Hell's Kitchen recorded the highest number of transactions (50,735), followed by Astoria (50,599) and Lower Manhattan (47,782). This confirms strong customer flow and stable business activity across all store locations.

Product type analysis revealed that:

- Barista Espresso generated the highest revenue (91,406.20)
- Brewed Chai Tea generated (77,081.95)
- Hot Chocolate generated (72,416.00)

These findings highlight the importance of specialty beverages in driving overall business performance and customer demand.

The analysis also showed that the average order value was approximately 4.69, indicating that customers generally purchase small-value items per transaction. This suggests that the business model depends more on maintaining a high transaction volume rather than relying on expensive individual purchases.

Another important finding was the relationship between unit price and quantity sold. The analysis showed that lower-priced products were sold in larger quantities, while higher-priced products were purchased less frequently. This demonstrates a strong relationship between product pricing and customer demand and highlights the importance of maintaining a balanced pricing strategy to maximize both sales volume and profitability.

Overall, the analysis successfully transformed raw transactional sales data into meaningful business insights. The findings can help coffee shop businesses improve product strategies, optimize store operations, enhance customer satisfaction, and support data-driven business decision-making.

5.6 FINAL INSIGHTS

- The business heavily relies on Coffee and Tea categories, which contribute the highest sales and quantity sold, making beverages the backbone of the coffee shop business.
- Coffee products generated the highest revenue of 269,952.45, followed by Tea with 196,405.95, showing that focusing on beverage-based products can significantly drive business growth.
- Sales are highly concentrated in top-performing store locations such as Hell's Kitchen (236,511.17), Astoria (232,243.91), and Lower Manhattan (230,057.25), showing the importance of location-based customer demand patterns.
- Morning hours between 8 AM and 10 AM are the most critical for business performance, with 10 AM alone generating the highest revenue of 88,673.39. This highlights the strong impact of breakfast demand and office-going customers.
- Monthly sales analysis shows strong growth in May (156,727.76) and June (166,485.88), indicating increasing customer demand and improved business performance during later months.
- However, sales drop significantly during evening hours, especially at 8 PM where revenue is only 2,935.64. This indicates the need for better evening sales strategies and promotional activities.
- Product analysis reveals that top products like Sustainably Grown Organic Lg (21,151.75), Dark Chocolate Lg (21,006.00), and Latte Rg (19,112.25) contribute significantly to revenue, while some categories like Packaged Chocolate and Flavours generate very low sales.
- This highlights the need for better product optimization, inventory planning, and targeted promotions for low-performing categories.
- Customer purchasing behavior shows that the average order value is only 4.69, meaning most customers purchase small-value items. This indicates that the business depends

more on high transaction volume rather than large individual purchases.

- Weekday sales are much higher than weekend sales, showing that regular working-day customers contribute more to overall revenue than weekend visitors.
- The analysis of unit price versus quantity sold reveals that lower-priced products are sold more frequently, while higher-priced products are purchased less often. This highlights the need for balanced pricing strategies to maximize both demand and profitability.
- Overall, the coffee shop demonstrates strong sales performance, but improving pricing strategies, strengthening low-performing product categories, and optimizing sales during non-peak hours can further enhance business growth and long-term sustainability.

CHAPTER 6: FUTURE SCOPE AND CONCLUSION

Conclusion

From this analysis, it is clear that data plays a crucial role in understanding business performance and making better decisions in the coffee shop industry.

The Coffee Shop Sales dataset reveals that sales are not equally distributed across all store locations, product categories, and time periods. Store locations such as Hell's Kitchen, Astoria, and Lower Manhattan contributed significantly to total revenue, showing the importance of location-based performance analysis.

One important observation is that Coffee and Tea dominate both sales and quantity sold, indicating that beverages form the core of the business. At the same time, products like Sustainably Grown Organic Large, Dark Chocolate Large, and Latte Regular stand out as the top revenue-generating products, making them key focus areas for business growth.

Another interesting insight is the presence of strong time-based sales patterns, with sales peaking during morning hours between 8 AM and 10 AM, especially around 10 AM. This suggests that businesses should focus more on staffing, inventory management, and promotional strategies during peak hours.

It was also observed that weekday sales were much higher than weekend sales, showing that regular working-day customers contribute more to overall revenue than weekend visitors.

The most important finding from this project is the relationship between pricing and customer demand. Lower-priced products were sold in higher quantities, while higher-priced products were purchased less frequently. This clearly shows that pricing strategy plays a major role in customer buying behavior and overall sales performance.

Overall, the analysis highlights that business success is not only about increasing sales, but also about understanding customer behavior, managing pricing strategies, and improving operations through data-driven decision-making.

Future Scope

- Apply Machine Learning models to predict future sales and customer demand
- Build interactive dashboards using Power BI or Tableau for better visualization
- Perform customer segmentation analysis for personalized marketing strategies
- Analyze product-level profitability in more detail
- Use real-time sales data for continuous business monitoring and decision-making
- Develop forecasting models for inventory planning during peak hours and high-demand periods

7. BIBLIOGRAPHY

- Pandas Documentation – <https://docs.python.org/3/>
- NumPy Documentation – <https://numpy.org/doc/>
- Matplotlib Documentation – <https://matplotlib.org/stable/contents.html>
- Seaborn Documentation – <https://seaborn.pydata.org/>
- Coffee shop sales Dataset - <https://www.kaggle.com/code/ahmedabbas757/coffee-shop-sales/input>

CONCLUSION

This internship has been an incredibly enriching and valuable learning experience that significantly strengthened my practical understanding of data analytics, business intelligence, and their real-world business applications. Through hands-on work with the Coffee Shop Sales dataset, I developed a deeper understanding of how data can be effectively used to analyze sales performance, understand customer purchasing behavior, evaluate product demand, and monitor overall business growth across different store locations and product categories.

Working with a structured transactional sales dataset helped me improve my technical skills in data cleaning, preprocessing, exploratory data analysis (EDA), feature engineering, visualization, and dashboard development. I gained practical experience in transforming raw transaction data into meaningful insights by analyzing trends related to store performance, product categories, customer buying patterns, sales timings, and revenue generation. This process enhanced my ability to work with real-world datasets, handle data-related challenges, and generate valuable business insights using analytical techniques.

Throughout the internship, I worked with various tools and technologies including Python, Pandas, NumPy, Matplotlib, Seaborn, Excel, SQL, and Power BI. These tools helped me understand the complete workflow of a data analytics project, from data collection and preprocessing to analysis, visualization, and reporting. Developing an interactive Power BI dashboard further improved my understanding of business intelligence and visual storytelling techniques.

Beyond technical knowledge, this internship greatly improved my analytical thinking and problem-solving abilities. I learned how to interpret patterns in data, identify factors affecting sales performance, and connect analytical findings to practical business situations. For example, understanding peak sales hours, identifying high-performing product categories such as Coffee and Tea, analyzing store-wise performance, and studying the relationship between product pricing and customer demand helped me understand how data-driven insights support business decision-making and operational improvements.

Additionally, the project strengthened my communication and presentation skills. By presenting insights through visualizations such as bar charts, line charts, heatmaps, pie charts, scatter plots, and interactive dashboards, I learned how to communicate complex analytical findings in a simple, structured, and understandable manner. This helped me improve my confidence in

presenting reports, explaining technical concepts, and sharing business insights with both technical and non-technical audiences.

The internship also taught me the importance of accuracy, consistency, and attention to detail while working with business data. I realized that effective data preprocessing and proper analysis are essential for generating reliable insights and making informed business decisions.

Overall, this internship has reinforced my interest and passion for the field of data science, analytics, and business intelligence. The practical exposure, technical knowledge, and project experience gained during this internship have equipped me with strong foundational skills, professional confidence, and a better understanding of how analytics can solve real-world business problems. This experience will play an important role in supporting my future learning, career growth, and contribution in a data-driven professional environment.

REMARKS

This is to certify that Mr. Sayyed Muhammed Shammass has successfully completed a two-month internship at Austrix Technologies LLC. During this period, he was associated with our team and contributed meaningfully to the tasks and projects assigned to him.

He displayed excellent enthusiasm and a strong desire to learn and grow. He consistently approached his responsibilities with diligence and professionalism. His ability to quickly grasp new concepts and apply them effectively was evident in the quality of work he delivered.

He collaborated well with colleagues, communicated clearly, and was receptive to feedback demonstrating a commendable attitude towards continuous improvement. His performance during the internship reflected a good balance of technical competence and a positive work ethic.

We believe that He possesses the qualities required to excel in his future academic and professional journey. It was a pleasure having him as part of our team, and we extend our best wishes for his continued success.

AYSHATH LUBNA

Internship Program Head

The first part of the paper discusses the importance of the research and the objectives of the study. It then presents a literature review of the existing research on the topic. The second part of the paper describes the methodology used in the study, including the data collection and analysis techniques. The third part of the paper presents the results of the study, and the fourth part discusses the conclusions and implications of the findings.

The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables.

The results of the study show that there is a significant positive relationship between the variables. This finding is consistent with the previous research on the topic. The study also found that there are some differences in the results between the different groups of participants.

The conclusions of the study suggest that the findings have important implications for practice. The study also identifies some limitations and suggests areas for further research.